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ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAILS IN GMAIL USING DECISION TREE ALGORITHM

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College of Computer and Information Sciences
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Bachelor of Science in Computer Science

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CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this thesis, **ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAILS IN GMAIL USING DECISION TREE ALGORITHM** for the degree Bachelor of Science in Computer Science at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This thesis does not contain words or ideas taken from published sources or written works that have been accepted as the basis for the award of the degree from any other higher education institution, except where proper referencing and acknowledgement of work were made.

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ABSTRACT

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The study "ePhishSense: Browser Extension Detection System for Phishing Emails in Gmail Using Decision Tree Algorithm" focuses on enhancing phishing email detection on Gmail platform using a browser extension employing the Decision Tree algorithm. Researchers aimed to mitigate fraudulent activities by analyzing the accuracy, precision, and recall of the system based on email features such as sender's address, DKIM, and URLs. Comparisons with the existing tool MightyG8 were conducted using a dataset of 1,000 emails from Bot Tool and GitHub, split into 80:20 ratio for training and testing respectively. The evaluation involved a sample of 200 emails, comprising both phishing and legitimate ones. Results showed ePhishSense achieved 98% accuracy, with 98.9% precision and 96% recall for phishing detection. Statistical analysis including a t-test (p-value = 0.026) demonstrated a significant improvement over MightyG8, supporting the efficacy of ePhishSense in detecting phishing attempts.

Keywords: Phishing, Browser Extension, Gmail, Detection, Decision Tree

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Phishing attacks have a significant global impact on users, particularly through the risk of data breaches, where cybercriminals gain unauthorized access to sensitive information stored within organizations. According to Griffin (2023) in “The latest 2023 Phishing Statistics,” published on February 06, 2023, phishing attacks affected a significant number of internet users globally in 2021, with 323,972 individuals falling victim to such attacks. In the first half of 2022, the Philippines witnessed a significant surge in phishing attacks as compared to the entire year of 2021. Over 1.8 million attacks were detected in this period, surpassing the total number of attacks detected in 2021, which was 1.34 million. The majority of these attacks were aimed at payment systems, e-commerce platforms, and local banks (Statistica Research Department, 2022). This indicates that phishing constituted a substantial portion of cybercrime incidents, accounting for half of the victims.

Despite Google’s robust cybersecurity measures successfully blocking 99.9% of phishing attempts, cybercriminals managed to steal a staggering $44.2 million through phishing attacks, with an average loss of $136 per attack. Emails remain the primary channel for phishing attacks, with an average of 16.5 leaked emails per 100 internet users worldwide in 2021. The most prevalent malware links found in phishing emails in 2021 were Trojans from the ‘Agensla family. These steal login credentials stored in browsers and credentials from emails. Phishing is considered the most disruptive form of cybercrime for UK businesses in 2022, tied with threat actors impersonating the organization online. Millennials and Gen-Z internet users (18-40-year-olds) are most likely to fall victim to

phishing attacks – 23% compared to 19% of Generation X internet users (41-55-year-olds).

Real-time protection and website monitoring make browser add-ons useful for phishing detection. These extensions analyze URLs, page architecture, and embedded scripts to swiftly identify and warn users about dubious websites or links, preventing phishing attacks and protecting important data. They also use phishing databases and reputation checks to compare web pages to known phishing sites, increasing internet security. According to Mehrabi et al. (2020), phishing email-detection browser extensions have grown in popularity as a means of protecting consumers from online schemes. These extensions function by comparing the content of incoming emails to a database of known fraudulent emails. If any suspicious content is detected, the extension can alert the user immediately and block access to potentially hazardous links or attachments. This provides an additional layer of protection against spoofing attacks and can assist users in keeping secure while surfing the web and monitoring their email. As of September 2021, StatCounter GlobalStats reports that Chrome has 64% of the global web browser market. Chrome's popularity led researchers to design a Chrome extension-based detection mechanism. A Chrome extension detection method can reach more Gmail users and improve their email security. The Chrome browser's large user base means that the developed method can protect many people from phishing assaults directly in their preferred browser. Additionally, based on Oberlo Statistics (2023), Safari, Edge, Opera and Firefox are widely used web browsers by users aside from Google Chrome. Also, Boyle & Shepherd (2021) found that browser extensions can be used as simple security tools that can be integrated into online activity. Browser extensions are simple and embedded, making them useful for internet novices.

Effective phishing attack detection and classification are crucial to cybersecurity because it protects people and businesses from criminal activity. Using machine learning algorithms, which have been proven effective in identifying patterns and making conclusions based on data analysis, is one well-known strategy in this field. The decision tree method has drawn the most attention since it is easy to understand, can evaluate the value of features, and can handle both category and numerical data. A decision tree algorithm is a valuable tool for finding and categorizing pertinent features connected to phishing attacks in the context of our study on phishing, assisting in creating efficient detection and prevention systems. One may learn more about the underlying patterns and characteristics of phishing assaults by comprehending the reasoning behind the decision-making process, leading to improved cybersecurity procedures. According to Abdulrahman & Salim (2022), emails are the cheapest and fastest way to communicate. However, the rise in email subscribers has drastically boosted spam emails. When classifying emails, the Decision Tree algorithm has higher F-measures, Precision, and Recall scores than the other algorithms. This study also showed the email spam algorithm to be the best. This study's recommendations can be applied to any email system worldwide because language stop word removal is included in email data pre-processing. Future research will examine various pre-processing, explore additional features, and feature encoding components to increase classification accuracy.

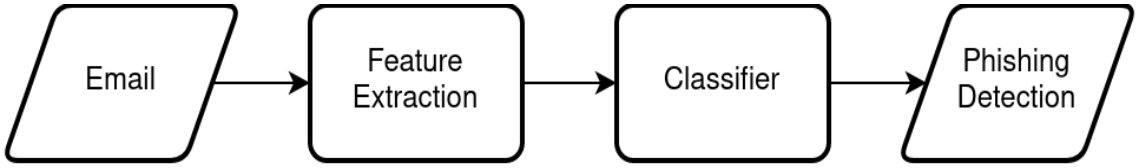
Additionally, email characteristics that have been found to be used in further studies involving other algorithms, but are not commonly used or lacking in research studies related to Decision Tree Algorithm may be used as phishing indicators that may be relevant to the said algorithm. Furthermore, Zhang (2021) pointed out that there were many uncommon features that are phishing indicators but not mentioned, and that additional research is needed to properly comprehend it. With this, the researchers will

take advantage of it and come up with the opportunity to test the accuracy of these email characteristics or phishing indicators if it is capable of producing acceptable accuracy.

This study was inspired by a firsthand experience with the terrifying implications of a phishing attempt, in which one of the researchers came perilously near to losing access to their personal account. This direct experience highlights the seriousness and real-world consequences of phishing attacks, providing strong reason to dive deeper into the complexities of such cybercrimes. In this research, the researchers deviate from Calvadores and De Vera's (2018) study on "Detecting Phishing Emails Using the O(ND) Algorithm." Instead, the researchers employ the Decision Tree Algorithm to evaluate its accuracy and effectiveness in detecting phishing emails. The researchers intend to develop a browser extension capable of detecting fraudulent phishing emails in the Gmail platform. The extension will analyze the content of these emails to determine whether they are fraud attempts or legitimate. The effectiveness of this extension in detecting fraudulent emails will also be evaluated.

Theoretical Framework

Figure 1. Phishing Detection Model by Calvadores & De Vera (2018)



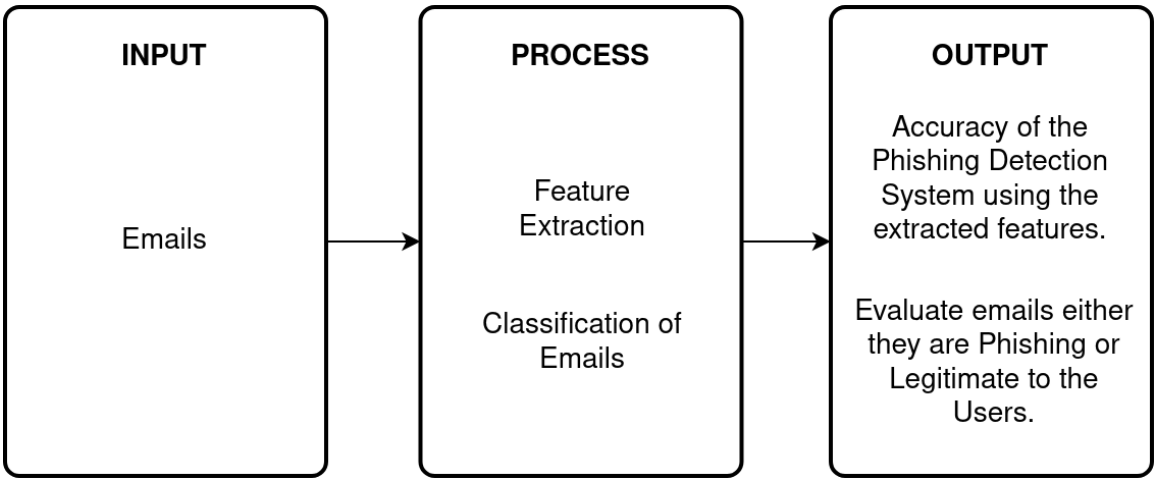
The model shows a Phishing Detection Model that has been used in the study presented by Calvadores & De Vera (2018). The said model includes four phases which consists of Email data, Feature Extraction, Classifier, and a Phishing Detection System.

For the first part, the input data would be a group of Emails collected and gathered by the researchers. These email messages are the data that needs to be classified

whether it is a legitimate email or a phishing one. Next, the Feature Extraction. It refers to the process of extracting or getting relevant information called features out of the raw email data. This would include information such as the header, subject, included links and other essential data that may be used for phishing detection. Then, the third phase is the classifier. A classifier is responsible for distinguishing between legitimate or phishing emails based on the extracted features or characteristics from the email data. Lastly, the phishing detection system itself. This phase refers to the final output of the model. It indicates to the user whether the given emails are legitimate or a phishing email based on the outcome provided by the classifier. With this, the said model will be used by the researchers as a reference model in developing their own phishing detection system.

Conceptual Framework

Figure 2. Conceptual Framework



The diagram above illustrates the conceptual framework of the study. The phishing detection system will utilize a Decision Tree Algorithm. The framework has three components: input, process, and output.

The system inputs both bot-generated emails and datasets containing labeled examples of phishing and legitimate emails. The system will use the datasets to train and evaluate the system.

For its process, Feature extraction was used to extract specific email characteristics that are possibly an indicator of a phishing email . These features include sender's email address, DKIM and URL/Outgoing Links. After that, the extracted features would be evaluated using a Classifier. For this study, the Decision Tree Algorithm has been employed to build a model using the labeled dataset. The model learns patterns and decision rules based on the extracted features. This trained model can subsequently be used to evaluate incoming emails and classify them as either "Safe" or "Malicious".

For its output, it measures the accuracy of the Detection System in correctly classifying phishing emails using the specified set of features. This accuracy is typically calculated by comparing the system's predictions to the known labels in the dataset. Then, it provides the classification of each incoming email as either "Safe" or "Malicious" to the users. This helps the recipient identify potential phishing attempts.

Statement of the Problem

The goal of the study is to propose a system that detects general emails sent in Gmail and determines its usefulness in protecting users from phishing emails. The study aims to answer the following questions:

1. What is the performance of the system in classifying an email as phishing or legitimate in terms of the following:
    a. Accuracy
    b. Precision

c. Recall

2. Is there a significant difference in the accuracy between the two studies, "Detecting Phishing Emails Using the O(ND) Algorithm" and "ePhishSense: A Browser Extension Detection System in Gmail for Phishing Emails using Decision Tree Algorithm," considering both studies utilized same email features for phishing detection?
    a. Sender's email address
    b. Domain Key Identified Message (DKIM)
    c. Outgoing links (URL)

Hypothesis

H_0: There is no significant difference in the accuracy between the researchers' study and the study with the system of Detecting Phishing Emails Using the O(ND) Algorithm.

Scope and Limitation of the Study

The study titled "ePhishSense: A Browser Extension Detection System for Phishing Emails in Gmail Using Decision Tree Algorithm" will focus on the development of a system that can distinguish between legitimate and phishing emails. ePhishSense is a browser extension-based system that is only compatible with web browsers on desktops and laptops, and Gmail users.

The Decision Tree algorithm will be used to determine whether an email is phishing or not. The tool is designed to scan the email when the extension was installed in the browser and upon opening the extension, the user must input their login credentials such as gmail address and app password, and the desired email subject name. The system will

only scan emails on the Inbox of Gmail. To ensure precise system scanning, emails must have a unique subject name. Lack of a subject name prevents the system from retrieving any data, and duplication restricts scanning to only the latest instance of the duplicated subject name. All emails will undergo the data de-identification process, which removes personal identification for data privacy.

The system will extract the Sender's email address, DKIM and URL/Outgoing Links, as these are the features of an email that will be used to further determine if it is phishing or not. The blacklisted URLs will be from the OpenPhish website that will be stored in CSV. The system will then determine if the email is malicious or safe by comparing it to the trained datasets and will send alerts to the user, whether it is malicious or not.

Significance of the Study

The significance of this study aims to create a browser extension that can identify phishing emails in Gmail. The said system protects users from online phishing tactics by providing a layer of protection against fraudulent attacks.

The results of the study will be a great benefit for the following:

Users/Community. The proposed study holds significant value for users and the community at large, including individuals such as general internet users, students, employees, online shoppers, and especially elders. By developing an effective browser extension, the study aims to empower them with enhanced protection against theft of personal and financial information, as well as fraud and phishing attempts. It can aid to prevent potential security breaches, financial loss, and reputational damages to users.

E-commerce. Subscribers can improve their online security and establish trust in their platforms by employing and commercializing detecting phishing emails in Gmail.

Institutions. The concept of a phishing detection system seeks to increase the accuracy of detection by employing a Machine Learning Algorithm, which holds crucial importance for institutions, especially those who focus on cybersecurity. The Decision Tree approach detects traits and attributes that are typically seen in phishing email.

Future Researchers. The findings and recommendations of the proposed study could serve as a basis for future researchers who want to study the field of phishing detection systems by providing citations for spotting phishing emails using a decision tree algorithm.

Definition of Terms

Blacklisted URL. It refers to a web address that is restricted or denied by specific systems, applications, or services.

Browser Extension. The type of software in how the system would be. It would be installed inside a Web Browser.

Comma-Separated Values file (CSV). A file that allows data to be saved in a tabular format.

Decision Tree. An algorithm that will be mainly used to classify email inputs whether they are phishing or legitimate based from the extracted email features.

Email Header/Footer. It contains information about the sender, recipient, subject, and routing information of the emails.

Electronic Mail Extension. A file extension used for email messages.

Feature Encoding. Refers to converting extracted feature values to a numerical value.

Feature Extraction. Is the process of getting relevant email characteristics that will be used to classify emails.

Phishing. A technique for attempting to acquire sensitive data.

Phishing Emails. A malicious email that is used to trick users.

True Positive Rate. The rate of how many emails has been successfully evaluated as either Phishing or Legitimate as correct by the system.

Uniform Resource Locator. It refers to the address of a given resource on the Web.

CHAPTER 2

REVIEW OF LITERATURE AND STUDIES

In this chapter, the researchers will explore and review various literature and studies that are relevant to the research topic of detecting phishing emails using a machine learning algorithm, specifically decision tree by examining research articles, journals, and other theses to define and support the research.

Phishing

As noted by Shankar et. al (2019), phishing is a type of cybercrime that tricks users into sharing confidential and sensitive information with attackers. This information can include credit card details, usernames, passwords, and bank account information. Phishing attacks can happen through malicious emails, text messages, and phone calls. Once the attacker obtains the information, they can use it for financial gain or identity theft. Phishing attacks can target individuals, organizations, or even specific groups within an organization.

Phishing attacks typically consist of three components: the lure, hook, and catch. Chaudhry, J. A (2016).

- The lure is usually an email that masquerades as a legitimate message from a reputable organization, like a bank or internet service provider, and includes a link to the hook.
- The hook, which often uses URL obfuscation to deceive the victim, is a fake website that looks like the legitimate institution's site and tricks the user into providing confidential information.
- The catch is when the attacker obtains and misuses the victim's sensitive data.

Additionally, they added that social engineers, a group of people who exploit other peoples' weaknesses. use a combination of tactics, including leveraging curiosity, fear, and empathy, as well as traditional phishing techniques, to deceive and manipulate users into becoming victims of phishing attacks.

Phishing is a deceptive practice that follows a specific sequence of steps. These steps include:

1. The attacker sends a fraudulent email, text message, or other type of communication to the victim.
2. The victim is enticed to click on a link or open an attachment in the message, which takes them to a fake website that looks like a legitimate one.
3. The victim is asked to enter sensitive information, such as login credentials, credit card numbers, or personal data.
4. The attacker collects the victim's information and uses it to gain unauthorized access to websites or conduct fraudulent activities.

Phishing attacks are often successful because they exploit the victim's trust in a seemingly legitimate source, as well as their curiosity, fear, or empathy. In some cases, the phishing email or website may also download malware onto the victim's computer, further compromising their security. (Bhavsar et. al, 2018)

Impacts of Phishing

As stated in the study of Sivakumar et. al (2023), phishing can have a devastating impact, particularly through identity theft. This occurs when cybercriminals use stolen personal information to commit fraud, resulting in significant financial losses and emotional distress for the victim. Identity theft involves the unauthorized use of another person's

identity for monetary gain or other advantages. In some cases, the perpetrators may even impersonate the victims themselves for fraudulent purposes.impact

In addition, according to Mohd Zaharon and Mohd Ali (2021), individuals should exercise caution and be aware of the social engineering tactics used by fraudsters to deceive their victims. Users must be suspicious when receiving calls, messages, or emails that create a sense of excitement or urgency, such as claims of winning a large sum of money or receiving a warrant from an authority. Failure to do so may result in falling victim to phishing attacks. Phishing attacks have several known consequences, such as the theft of personal information, including financial data or fraudulent credit card charges. Additionally, attackers may steal digital data, such as documents, files, and media. They may also steal identities, including social media posts or messages sent from the victim's account, often as part of further phishing attempts.

Phishing attacks can have devastating consequences, not only financially but also emotionally. The victims may find it difficult to trust anyone online, which can cause significant mental stress. This emotional distress may not only stem from the stolen information itself, but also from the ways the phishers may use the information. These criminals may even use email threats to further intimidate the phishing victim. In extremity, victims may not only experience anger and fear, but also develop Post-Traumatic Stress Disorder (PTSD). (Mahamood et. al, 2023).

By correlating all of them, it is clear that phishing is a highly prevalent social engineering attack that has disturbed people's lives as well as badly impacted countless organizations by collecting sensitive information through various manipulation techniques. It has major impacts, including compromised personal data, financial loss, identity theft, data breaches for enterprises, malware infections, loss of confidence, reputational harm,

and possibly legal and regulatory ramifications. The increasing popularity of phishing detection software and plugins reflects the changing nature of these attacks and the necessity for effective defenses (Shah et al., 2020).

Email Features

This section describes the features that the researchers will utilize for email classification. Althobaiti et al. (2019) mentioned that they have observed that a wide range of features has been explored in the context of automated phishing detection. However, they have also identified certain categories of features that have received limited attention in research. Thus, a series of features typically employed by phishing attackers were identified and used in this work; the combination of these elements formed a feature set that effectively classified emails as phishing or non phishing.

The remainder of this section describes these features.

a. DKIM: SPF and DKIM are recommended by the CISA (n.d.) for detecting illegal emails. DKIM (DomainKeys Identified Mail) allows a transmitting domain to "watermark" their emails, making unauthorized emails (e.g., spam, phishing email) easily detectable.
b. Outgoing Links (URL): This attack involves malicious actors disguising the victim's desired web link (URL). Make their web address look credible. As Ollmann (2004) showed, these attackers generate bogus web addresses using dodgy domain names and complicated hostname tactics.

C.1. Blacklisted URLs

A blacklist is made up of URLs that violate these standards, indicating sites that are judged undesirable or non-compliant, as well as

> possibly phishing sites. The adoption of URL filtering acts as a quick and effective technique for regulating online traffic from undesired locations, functioning in a stream-based manner. This technique operates on a worldwide scale, allowing for the collective banning of traffic for all businesses on the blacklist or the selective allowance of traffic for URLs on the whitelist. When a whitelist match is found, the communication is not impeded and may be tracked as configured. A blacklist match, on the other hand, leads to the request being dropped immediately, preventing it from progressing to the intended destination and documenting the event if specified. When neither a whitelist nor a blacklist match is found, traffic is unaffected.(Allied Telesis, 2018)

Different Approaches on Phishing Detection

Strong security measures and attentive monitoring are necessary to identify and stop phishing attacks. Different strategies have been developed to recognize and lessen these hazards.

Natural Language Processing (NLP) is a different method for examining the content of emails or messages. Attackers modify their strategies constantly to go around improvements in defense and take advantage of newly identified flaws or circumstances. To identify the assaults, the current anti-phishing software combines blocklist, heuristic, visual, and machine learning methods. In contrast to looking for dangerous URLs and emails from a number of databases, Sahingoz et al. (2019) advocate employing classification algorithms and basic NLP characteristics. An accuracy rate was established in experiments using a newly created test dataset and the Random Forest model with NLP-based features.

The study categorizes and offers novel email filtering methods for detecting targeted malicious email (TME), which promotes computer network exploitation. TME is a malicious email attack unlike spam and phishing. Using persistent threat and recipient-oriented features, supervised data classification algorithms offer new TME detection methods. The classifier's performance is assessed and compared to conventional email filtering methods to show the increased advantage of these variables. The study reveals that persistent threat and recipient-oriented features dramatically reduce false negatives and acceptable false positives. The study can be expanded to include multi-class outcomes (2010, Amin).

Phishing attacks can destroy physical and digital systems, leak sensitive data, ruin reputations, and cost organizations money, according to Li et al. (2022). Deep ensemble learning analyzes all email segments, including attachments, in the authors' completely automated malicious email detection method. The framework's AUC of 0.993 outperforms state-of-the-art malicious email detection approaches, including human expert feature-based machine learning models, by 5%. This paper suggests a way to tackle complicated email-based threats like malicious email attachments without feature engineering.

The study suggests using email header information to detect phishing and spam emails without using the email body or subject. Features are extracted and selected using one-class and multi-class anomaly detection techniques. The study found that supervised learning algorithms such Random Forest, SVM, MLP, and KNN, as well as their stacked ensembles, have 97% phishing and 99% spam email accuracy. For spam and phishing emails, One-Class SVM obtained 87% and 89% accuracy. The results show that email header information alone may consistently detect these anomalies, making email filtering software more efficient and resource-efficient.

Basically, in a Rule-based approach, every email that would be checked by the Phishing Detection System using Rule-Based Algorithms would follow a set of rules or other predefined configuration. This may include verifying the header file or the content itself. Then from it, the system would decide whether the email is a phishing email or not.

As stated by Jamel et. al. (2013), phishing detection techniques rely on the identification of specific features, typically found in the email header and body. In the context of their study, a total of 18 binary features have been extracted. These features are extracted using the Visual Basic.Net programming language, enabling the analysis and processing of the relevant data. The selection and extraction of these features play a crucial role in the effectiveness of the phishing detection system, allowing for the identification of suspicious patterns and characteristics indicative of phishing attempts.

According to Yuan, Z., Yuan, Q., & Wu, J. (2020, November 12), with the extensive use of blockchain in the financial sector, blockchain security faces significant problems from cybercrime, particularly phishing schemes. On Ethereum, the article offered a graph-based categorization method. They gathered transaction logs from some validated phishing addresses as well as an equal number of legitimate addresses. Second, in order to represent the original address on a graph level, they created a series of subgraphs, each of which has a target address and its surrounding transaction network. Finally, they suggested an enhanced Graph2Vec and made classification predictions on the subgraphs they generated based on their study of the ether flow of the phishing scam cycle. The experimental findings reveal that the framework obtained a high level of competitiveness in the final classification test, indicating the potential utility of phishing detection on Ethereum via learning the representation of the transaction network.

According to Basnet (2012), The rule-based approach in detecting phishing attacks involves searching for specific patterns that is similar to a network Intrusion Detection System (IDS) signature, within a webpage. The goal of this approach is to create an intuitive and user-friendly framework that is also flexible and can be easily expanded to include new and emerging phishing tactics. Based on their study, the rule set is generated through tier observations and by utilizing machine learning features proposed in various existing literature on phishing attack detection. By continually updating the rule set, the detection system can adapt to evolving phishing techniques and improve the effectiveness of our detection methods.

The researchers advise improving the algorithm to detect phishing emails faster on slower PCs. Minimizing the tool's console output to focus on vital information optimizes processing efficiency. Upgrading to Gmail.js's current version improves efficiency and adds new features, improving the system's ability to detect phishing emails quickly (Calvadores & De Vera, 2018).

Machine learning

Machine learning is the most popular technique for identifying malicious and suspicious websites by using URLs. Classification of phishing URLs is an important domain in machine learning. A large number of data features are required to acquire machine-learning-based security systems and to train the model on features that are associated with legitimate and phishing website labels. The outstanding performance of machine learning algorithms allows them to easily detect hidden or first-time attacks that are not on a blacklist. (Abdul K. et al., 2023).

Semantic analysis of email text is used to identify and classify malicious emails in the study. Machine learning algorithms analyze email body and topic semantics to detect

malicious information. The method had 96.35% accuracy on a dataset of over 4,000 benign and malicious emails. In the face of targeted attacks, the approach achieved 94.32% accuracy. The proposed method can detect unknown or zero-day malware and strengthen email security systems. Real-time email filtering can use the computationally efficient method (Al-Saadi et al., 2022).

According to (Javed, 2020), a supervised learning problem in the field of machine learning is Email classification which involves the author prediction of an email based on the context of the text. The literature tests various machine learning algorithms for email classification and compares their performances including decision trees. In decision trees, running the code above gives us an accuracy of 0.9880 and a training time of 6.116 s. a very good accuracy score with 100% of the training data, but the best machine learning approach for email classification is governed by a variety of factors, including the kind of dataset, the size of the dataset, and the issue's particular requirements.

As stated by Pujara and Chaudhari (2018). Machine Learning approach has shown efficient results in detecting phishing attacks in large datasets, and it is capable of detecting zero-day attacks. Machine learning-based classifiers have achieved over 99% accuracy in identifying phishing websites. However, the performance of the classifiers depends on the size of the training data, feature set, and the type of classifier used. One limitation of this approach is its failure to detect compromised domains used by attackers to host their phishing websites.

On a side note, Rose, M. A. S. R. et. al (2022) also used a self-destruct detection algorithm to propose a machine learning model that is then used to develop a chrome-based phishing detection system. The said system is expected to provide necessary countermeasures and preventive features to keep the user from accessing malicious websites while relying mostly on URL-based features.

Detecting phishing websites through a comparative and analytical evaluation of various machine learning algorithms was conducted in this paper. The researchers analyzed a huge amount of data consisting of 6157 original websites and 4898 phishing websites using twelve various classifiers, which include logistic regression, decision tree, random forest, ada-boost, support vector machine, KNN, artificial neural networks, gradient boosting, and XGBoost. The authors concluded that machine learning is a powerful tool for detecting patterns in data and recognizing phishing websites. (Darabi et al., 2020)

Additionally, numerous studies have been conducted in the area of phishing detection, with most focused on improving the accuracy of detection using different classifiers. Several classifiers have been used, including KNN, SVM, Decision Tree, ANN, Naïve Bayes, PART, ELM, and Random Forest. Based on the literature survey, tree-based classifiers such as Decision Tree and Random Forest have shown to be the most effective, particularly with increasing dataset size.

Overall, the Machine Learning approach makes use of a large volume of dataset of known phishing emails that will identify its characteristics or other features that are usually found on phishing emails. It would provide an outcome whether the input email is distinguished as email or not.

Decision Tree Algorithm

Decision trees are considered as one of the most efficient and effective discrete methods for supervised learning, as they produce output in a graphical format. Decision trees do not require any explicit rules to be specified but instead require a pre-classified dataset as the training sample. This algorithm maps each element in the given domain to an element in its range, which can be either discrete or continuous. Decision trees are

particularly useful for class-type variables. In this process, each split is chosen in a way that it reduces the target variable's variance. (Unnithan et. al, 2018).

Additionally, they added that the input to the decision tree is usually passed as an object or scenario that simulates a set of properties, and the output is typically a decision that can be either YES or NO. The decision tree is built using leaves, and at each node of the tree, a test is conducted to find the least possible outcome. The leaves consist of numerical or categorical values, which represent the outcome after each test.

Based on the study of Mahajan R. and Siddavatam I. (2018), the decision tree algorithm is a popular technique in machine learning due to its simplicity and ease of implementation. It starts by selecting the best attribute to use as a splitter for classification, which becomes the root node of the tree. The algorithm continues building the tree until it reaches a leaf node. The resulting training model can then be used to predict the target value or class. In the tree representation, each internal node corresponds to an attribute, and each leaf node corresponds to a class label. The decision tree algorithm utilizes the Gini index and information gain methods to calculate these nodes.

According to Bhandary Prajwal Gopal Krishna, D. S. Rajesh, and K. Prasganth Kumar (2021) define decision trees as a supervised learning technique that can be applied to classification and regression problems, although it is most commonly used for classification problems. Their study uses the decision tree algorithm to represent the features of their datasets, which consists of two nodes: the decision node, which is used to make numerous decisions, and the leaf nodes, which are the results of these decisions. Their research used the Random Forest Algorithm, which can be applied to classification and regression issues. Instead of relying on a single decision tree, this algorithm extracts

the predictions from each tree and bases the final output on the greatest number of predictions.

In light of this study, Ahamed et al.'s (2022) research nearly follows the same methodology. They classify the data based on its features using machine learning algorithms such as random forest classifiers and support vector machines. The applied model is a decision tree. It is trained by supplying both the original and phishing links so that it can provide correct accuracy. The overfitting of decision tree classifiers is one of the authors' main worries, but decision trees successfully classified their training data set but gave poor results with a testing dataset. In addition, they noted that this algorithm provided the highest classification accuracy in their research.

According to Kumar et al. (2021), decision trees are a form of Supervised Machine Learning in which data is continuously split based on a particular parameter. According to them, the most difficult aspect of a decision tree is identifying the attribute for each root node. Information Gain and the Gini Index are two of their most popular attribute options. Their study employed three machine learning techniques, namely Random Forest, Logistic regression, and Decision tree. Their study found that Decision trees have the lowest predictive accuracy and existing technique accuracy. They added that classification performance is enhanced when more data is used as training data and they recommended that their proposed system can be improved by increasing the dataset and creating a browser extension.

Genetic Decision Tree Processing with Natural Language Processing (GDTPNLP) is used to process text- and voice-enabled emails. Email spam can be identified in textual and speech-enabled emails using this method. The GDTPNLP technique improves spam detection speed, performance, cost efficiency, and accuracy. For future work, improve the Browser extension detection system's time efficiency. Data pretreatment and feature

extraction should be prioritized. N-Complex optimization techniques are efficient at managing huge datasets. The technology can be optimized for real-time applications to detect phishing emails faster and more accurately (Ismail et.al., 2022).

Browser Extension

According to Pavan, M., Venkata Sai Mounika, B., & Mahitha, G. (2023), a Chrome browser extension or plugin can be a valuable tool for integrating phishing detection into the user's surfing experience and providing real-time alerts when a suspicious link is clicked. The implementation of a chrome-based plugin also makes it easier for users to identify phishing websites because they no longer need to rely on traditional approaches such as maintaining and updating a blacklist database. However, because the plugin is working with sensitive data, it must be trustworthy and safe. Further review and testing of the plugin would also be desirable to confirm its effectiveness and correctness in real-world circumstances. The employment of a machine learning algorithm with a chrome-based plugin has the potential to significantly improve phishing detection and reduce the financial and economic harm caused by phishing attacks.

According to Chen and Hossain (2021), for a variety of reasons, the Google Chrome extension is an excellent development platform. To begin, the Google Chrome Extension is based on and installed within the Chrome browser, which has a minimum impact on the use of computer system resources. Chrome is a very popular browser all around the world, and people may simply access it. Second, because most people see and send emails through browsers, the browser's built-in functions, like extensions, will be the most basic and effective way to detect phishing emails. Third, the extension will be small and simple to install within the browser. The Google Chrome extension is safe and

simple to use because it only requires one click to install. Users can disable the extension with a single click if they want to discontinue it.

In addition to that, Gurjar et. al. (2021) proposed a security model namely “WebSectAsst.” It is a browser extension that uses Machine Learning Model to distinguish possibly unsafe web pages that are visited by the users. Furthermore, they included the capability to determine whether the direct downloads from that website may expose certain risk or if it is a legitimate download. Lastly, they opt to highly value the privacy of the user’s data thereby claiming that the said software preserves those data rather than the available chrome plugins.

Related Studies

According to Chakraborty and Mondal (2012), machine learning-based filtering techniques can be categorized into two types: complementary and complete solutions. Unlike rule-based approaches, machine learning approaches do not require the explicit specification of rules. Instead, they rely on a set of pre-classified documents as training samples, which are used to train a specific algorithm to learn the classification rules. This approach is highly effective and can improve the accuracy of phishing email detection.

Since used bottles pollute water and land, Bottle Recycling Machines (BRMs) help solve solid waste issues. The researchers suggest a new BRM design to gather used bottles and categorize them using machine learning. After sorting, used bottles are recycled separately. The system includes raspberry-PI, camera, and audio-visual interactive system. Thermal receipt printer coupons are used to reward users. Since there is no sensor to detect plastic bottles, machine learning algorithms are used. Bottle purchase is first. Images and extracted features are submitted to novel classifier (machine

learning algorithm). BRM's proposed design accurately identifies bottles and recycles them cheaply (Dhulekar et. al., 2018).

According to Feng (2022), due to the global network economy, global market competitiveness has increased, commercial project scale and number have grown, and human resource planning and management requirements have increased. This article studies the human resource intelligent scheduling algorithm based on the decision tree algorithm, analyzes it in detail, finds the core problem, integrates human market resources, and solves modern human resource management's key problems. It references the decision tree algorithm in human resource intelligent scheduling. Human resource scheduling using the decision tree method cuts project cycle time by 27.5%, according to experiments. The experiment's human resource scheduling time savings are met.

This study analyzes decision tree (DT) methods for power system real-time dynamic security assessment (DSA). This paper covers three widely used DT algorithms: random forest, REP, and J48. Three decision tree (DT) methods are tested for power system network dynamic security assessment (DSA). Accuracy, inaccuracy, and computational time evaluate algorithms. PowerWorld program tests these methods on a modified IEEE 30-bus test system model. This study shows that random forest algorithm outperforms min technique in accuracy and error. REP's dynamic security assessment technique is faster than others (Al-Gubri et. al., 2016).

Highly reliable rotating machinery helps prevent unscheduled maintenance, which costs power plants. Vibration analysis helps improve rotating equipment reliability. Vibration data can identify equipment failure symptoms. Classifying vibration symptoms with one bearing angle has been done earlier. Bearing angles are horizontal, vertical, and

axial. To classify vibration symptoms, this study analyzes horizontal, vertical, and axial angles. This study classified using the Decision Tree technique in two phases. The first step classification employs seven parameters collected from vibration data, while the second step classification uses three parameters. In this study, all five instances were classified correctly in the first stage while imbalanced, parallel misalignment, and shaft bending cases were classified correctly in the second step. Decision trees can classify vibration symptoms, according to this study (IEEE, 2021).

Kutlucan et. al. (2023) stated that microfossils can be classified to learn about Earth, life, ecology, and biological changes. Traditional classification methods require paleontologists to carefully examine each microfossil sample under a microscope. It's pricey and takes a long time to process. Deep learning and shallow machine learning methods were used to classify Globotruncana species in this article. The deep model specifies data using high-level layers, while shallow models recognise low-level levels during learning. Researchers labeled more than 180 microscopic pictures of three Globotruncana species. After applying machine learning approaches to this dataset, the researchers analyzed the outcomes. Experimental results indicated that deep and shallow machine learning methods can classify Globotruncana species with up to 100% accuracy.

Research Gaps

While SVM, Random Forests and other machine learning algorithms were widely used in phishing detection, Decision Tree Algorithm may also provide better accuracy. Features of emails in other studies that weren't used by the Decision Tree Algorithm were used in this paper to determine if Decision Tree can determine the legitimate and phishing emails through these unusual features.

Synthesis of the Study

Phishing email detection is a critical aspect of web security, aimed at identifying and preventing fraudulent emails that attempt to deceive recipients into revealing sensitive information or engaging in malicious activities. Thus, the researchers ought to prevent this by developing a Phishing Email Detection System with the use of Machine Learning Algorithm. The Decision Tree has been widely used in the field of phishing email detection due to their ability to classify and differentiate between legitimate and malicious emails based on predefined rules and features. Along with the said algorithm, the researchers will use specified email characteristics which are used in other studies but are still not commonly used with the decision tree algorithm.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology employed in conducting the research on ePhishSense: A Browser Extension Detection System for Phishing Emails in Gmail Using Decision Tree Algorithm. It consists of the research design, sources of data, research instrument, data generation procedure, ethical considerations, software development, and the system architecture.

Research Design

The research design adopted for this study was an experimental approach. The experimental research design, specifically the post-test approach, involves the systematic manipulation of variables and the collection of data to test specific hypotheses and draw conclusions about the relationship between variables. The post-test approach is used in similar research context, as evidenced by its application in previous study on phishing detection entitled MightyG8 by Calvadores and De Vera in 2018.

In this case, the research design involved conducting experiments to evaluate the performance of the decision tree algorithm in detecting phishing emails. The experiments included training the algorithm on a labeled dataset of phishing and non-phishing emails, applying the algorithm, and measuring its accuracy, precision, and recall.

Sources of Data

The researchers collected 1,000 emails relating to the study. The target users of the detection system were those who have poor digital literacy, diminished cognitive capacities, and higher financial risk, especially those who are over 60 to 65 due to their vulnerability to phishing attacks and online scams. The obtained data were divided in an

80:20 ratio, with 80% of the data utilized for training the system and the remaining 20% used to evaluate its effectiveness.

Training Dataset

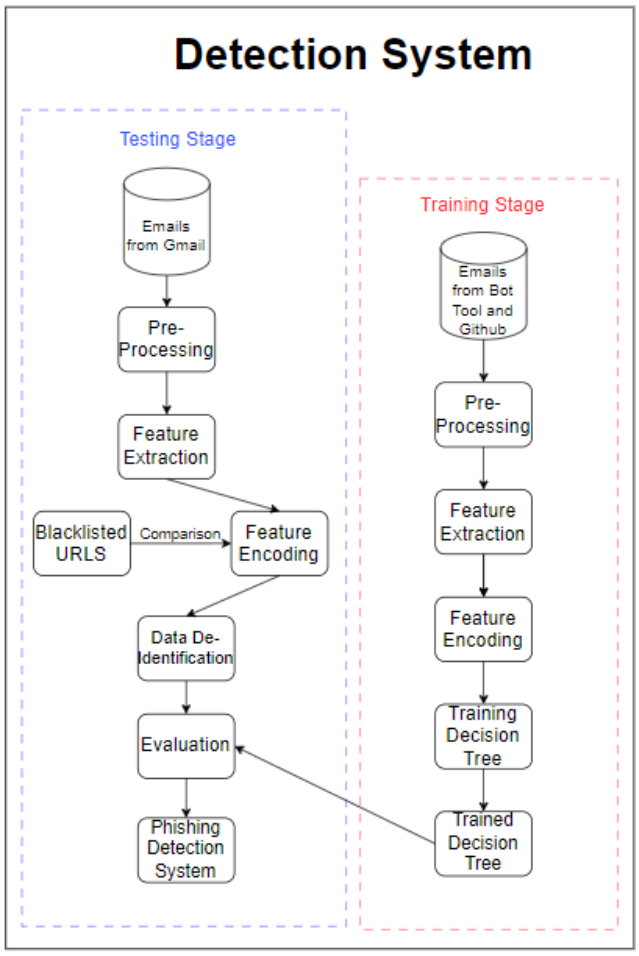
The main data source was GitHub and the Email-Bot Tool for training dataset. This dataset comprised 800 relevant emails. These emails were trained and evaluated by the decision tree algorithm.

Testing Dataset

The researchers generated emails from GitHub and the Email-Bot Tool, which was created by Calvadores & De Vera in 2018. The researchers generated 200 emails that were used for testing.

System Architecture

Figure 3. System Architecture



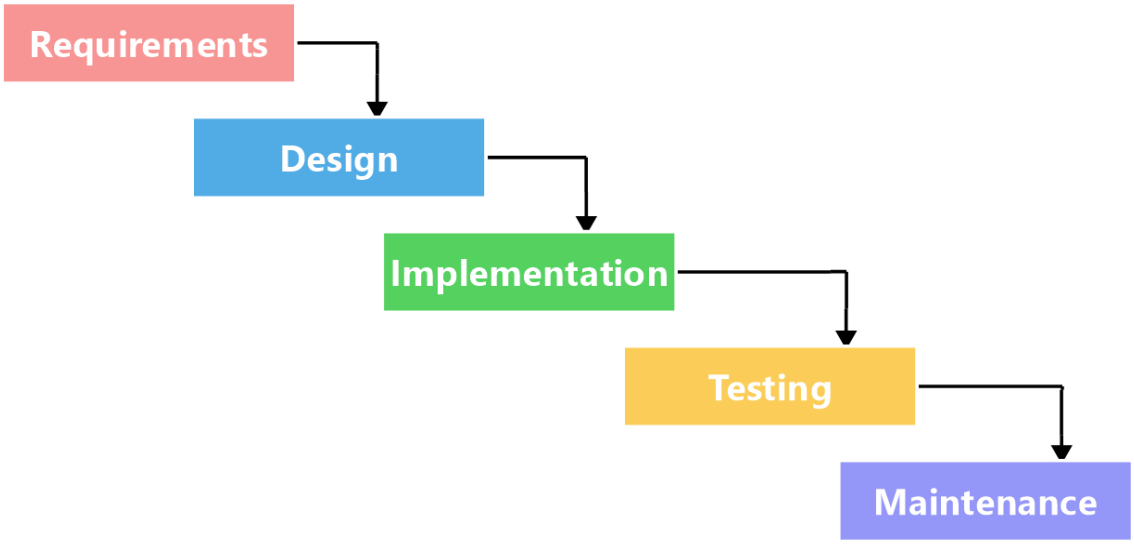
The system architecture of the detection system, as depicted in Figure 4, is as follows:

1. **Input**: After the user logged into their account, the system took emails as input from Gmail, scanning them from the Gmail platform through subject name after inputting the desired email from the inbox.
2. **Pre-Processing:** The system performed pre-processing tasks to clean and transform the raw data obtained from the email's content.
3. **Feature Extraction:** Relevant data was identified from the email using three features: Sender Email Address, DKIM, URL/Outgoing Links.

4. **Feature Encoding:** The extracted features were encoded into a numerical format that could be understood by the Decision Tree Algorithm.
    a. Sender’s email address was converted to numerical values such that 1 as present and 0 as none.
    b. DKIM was converted to numerical values such that 1 as present and 0 as none.
    c. URL was analyzed using URL comparison and was converted to numerical values such that 1 represented its presence, -1 represented being on a blacklist, and 0 represented none.
5. **Data De-Identification:** The system deletes the extracted data once it was converted into numerical values.
6. **Decision Tree Model:** The researchers collected a total of 1000 emails from Github and Bot Tool. These emails were then annotated based on features such as Sender's Email Address, DKIM, and URL, indicating whether each feature is present or not in the email. The researchers developed a program using a decision tree algorithm that trained on 80% of the dataset and tested on the remaining 20%. The program generated a model with an accuracy of 98%, precision of 98%, and recall of 96%. This model will be utilized to evaluate emails on the Gmail platform.
7. **Evaluation**: The system evaluated whether an email was considered phishing or legitimate based on the Trained Decision Tree Model.
8. **Output**: The system displayed the results to the user, which were Safe or Malicious.

Development Details

Figure 4: Waterfall Framework



The researchers developed the system following the Waterfall Framework. According to Finlay (2021), the waterfall methodology is often regarded as the traditional method of project management. When teams used it, they divided projects into linear, sequential steps. Researchers collected all needs at the start of the project and used this knowledge to develop a plan. The researchers developed a strategy for early planning that was fairly rigid; the next phase could not be started until the tasks in the preceding phase were completed.

The research required the implementation of an algorithm, which is Decision Tree, and the development of a detection system; a sequential and linear approach was the waterfall methodology. The waterfall approach led the general development process with its simplicity by following a step-by-step process, allowing the researchers to track progress with ease. The decision tree algorithm was designed, implemented, and tested as a part of an email detection system, enabling the researchers to enhance the accuracy and ensure the reliability of the phishing detection.

The researchers utilized Python, a dynamic and interpreted language that supports object-oriented programming. The frameworks and libraries of Python were used for training the decision tree model that was integrated into Visual Studio Code (VSCode), a popular and versatile Integrated Development Environment (IDE) where the phishing detection software was developed. The researchers used the strengths of these tools to successfully design the decision tree algorithm that created the Browser Extension Detection System and carried out the research efficiently.

Research Instruments

The study used the following tools in the development of the system.

Development Tools

To perform the process of the study, the researchers required required the following qualifications of software and hardware:

1. Desktop/Laptop - These were the hardware devices where the IDEs ran.
2. HTML - It was used to structure the content and layout of the extension's user interface.
3. CSS - It was used to style the HTML elements within the extension.
4. JavaScript - It was used to add interactivity and functionality to the extension. It was also used to manipulate the DOM, handle user events, and interact with the Browser Extension API.
5. Browser Extension API - It was an extensive API that allowed interaction with various features and functionalities of the browser.
6. Python - A general object-oriented programming language that will be used to implement the system.
7. Visual Studio Code - A text editor which provides many features making it easier to write and manage code and text.

8. Flask - It was a lightweight web application framework for Python.

Experiment Materials

The researchers chose datasets because it offered many benefits that supported this thesis. Datasets provided a structured and comprehensive collection of data for the ePhishSense tool in training, testing, and evaluation. By gathering a collection of phishing emails, the researchers exposed the system to a variety of phishing strategies and patterns, improving its ability to distinguish real emails from phishing attempts. The study used an experiment paper to record test results during the experimentation period.

Data Gathering/Generation Procedure

Email data was gathered through two different sources. The first dataset was coming from GitHub. The following were the steps that the researchers used to gather email data:

1. The researchers browsed through the website for the available email dataset that contained both phishing and legitimate emails.
2. The researchers downloaded the most related and recent phishing email dataset. Downloaded dataset mostly in ".eml" format.
3. The dataset was then used for feature extraction and classification.

Secondly, the email data was gathered from an Email-Bot Generator. The said bot generated a number of emails that were legitimate and phishing. The following were the steps that the researcher followed to generate emails with the use of the bot:

1. The researchers used an Email-Bot Generator developed by Calvadores & De Vera in 2018.
2. The bot generated random legitimate emails.
3. Lastly, the emails were collected and were subject to pre-processing and feature extraction.

Ethical Consideration

In conducting this research, ethical considerations were placed into the utmost importance to safeguard the privacy and confidentiality of individuals and organizations involved.

Informed Consent - the system provided a consent agreement for the user to sign before they could access the service. The said agreement displayed necessary information such as what the system did to the data collected, where the data was stored, in what form it was stored, what data was being collected, and other relevant information.

Data Privacy and Security - the system was expected to comply with standard data privacy rules and regulations for the data collected. Furthermore, it provided appropriate security measures to protect the information from unauthorized access, use, or disclosure.

Data De-Identification - the system went through data deletion after converting the extracted data into numerical values. This ensured that sensitive and personal information was not retained in its original form once it had been processed and transformed into a format that was less likely to be linked to an individual.

The researchers were committed to maintaining the confidentiality of the data and using it solely for the purposes outlined in the research objectives. Any findings or results

obtained from the analysis of the data were presented in an aggregated and anonymized manner to prevent the identification of individuals or organizations.

Data Analysis

Understanding data analysis is critical for both researchers and decision-makers, allowing them to identify patterns, extract meaningful insights, and make informed decisions. Effective data analysis not only converts raw data into actionable information, but it also ensures the consistency and credibility of research findings. According to Calvadores and De Vera (2018), a solid understanding of data analysis is critical for validating research findings and facilitating evidence-based decision-making.

Accuracy measured the overall correctness of the system's predictions. It was calculated as the ratio of the correctly classified instances (TP and TN) to the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measured the proportion of correctly classified positive instances (phishing emails) out of all instances predicted as positive. It indicated the system's ability to avoid false positives.

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

Recall measured the proportion of correctly classified positive instances out of all actual positive instances. It indicated the system's ability to detect true positives without missing any.

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

To calculate the following, the researchers considered the following:

- **True Positive (TP):** System correctly identified a phishing email as phishing.
- **True Negative (TN):** System correctly identified a legitimate email as legitimate.
- **False Positive (FP):** System incorrectly identified a legitimate email as phishing.
- **False Negative (FN):** System incorrectly identified a phishing email as legitimate.

These terms were used to assess the performance of the system and evaluate its accuracy, precision, and recall.

Table 1

Rating System for Accuracy of a System (Eboña, 2013)

Computed Value	Verbal Interpretation
97% - 100%	Excellent
91% - 96.99%	Superior
89% - 92.99%	Very Good
85% - 88.99%	Good
80% - 84.99%	Satisfactory
75% - 79.99%	Fair
70% - 74.99%	Pass
Below 70%	Fail

Table 1 shows the interpretation of the performance of the system. The rating system's acceptance is based on how well it matches the evaluation objectives unique to the ePhishSense system. The parameters that were chosen—Accuracy, Precision, and Recall—were carefully picked to evaluate the system's ability to distinguish between phishing and authentic emails. These metrics provide a systematic and exhaustive

method, guaranteeing an accurate and comprehensive assessment of the functionality of the ePhishSense system. The rating system, which was influenced by (Eboña, 2013), offers a reliable and pertinent framework that easily fits with the goals of ePhishSense. Based on the table, a percentage below 70% is considered as Fail. On the contrary, 70% and above percentage are considered to be within the acceptable range.

In conclusion, the data analysis in this study focuses on assessing the performance of a phishing email detection system using the decision tree algorithm. Researchers were able to assess the system's effectiveness by looking at critical metrics like accuracy, precision, and recall. These metrics provided clear, objective insights into the system's performance, allowing for a thorough understanding of its ability to detect and prevent phishing attacks.

Statistical Treatment

The statistical treatment of data is critical in research because it facilitates the systematic organization and interpretation of raw data. It entails analyzing data using appropriate statistical methods to support hypotheses and improve the rigor of research methodologies. Researchers can evaluate differences between groups and draw meaningful conclusions by using techniques such as the Independent Sample T-test (Equal Variance). The use of statistical tools like SPSS at a significance level of 0.05 (alpha, α) demonstrates the analytical process's precision and reliability.

T-Test

The study used the T-Test, specifically Independent Sample T-test (Equal Variance) formula to compute and compare the given results of the proposed study and

the Phishing Detection System Using O(ND) Difference Algorithm. T-tests was used to determine the significant difference between the two groups. The formula is given below:

$$t = \frac{a1 - a2}{\sqrt{\frac{s1}{x1} + \frac{s2}{x2}}}$$

Where:

a1 = mean of the first set of values

a2 = mean of the second set of values

s1 = standard deviation for first set of values

s2 = standard deviation for second set of values

x1 = total number of values in the first set of values

x2 = total number of values in the second set of values

Standard Deviation Formula

$$s = \sqrt{\frac{\Sigma (n - m)2}{x - 1}}$$

Where:

n = values given

m = mean

x = total number of values.

The level of significance used in this study was 0.05, denoted by alpha (α). The statistical tool used for computation of the significance was SPSS.

In conclusion, the statistical treatment used in this study provided a strong framework for data analysis, allowing researchers to draw conclusions and effectively

compare different models or approaches. These analytical methods contribute to a thorough understanding of the subject, resulting in strong conclusions in research endeavors.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter includes the data presentation, interpretation and analysis of the outcomes of the data collected by the researchers. It also presented the analysis on what the result means through representations of tables in response to the statement of the problem.

The researchers conducted experiments to evaluate the performance of the system in identifying emails on the Gmail platform as phishing or legitimate in terms of accuracy, precision, and recall.

The objective of this research is to lessen phishing attacks by developing a browser extension detection system that determines phishing and legitimate emails. The researchers aimed to solve the following research questions:

1. What is the performance of the system in classifying an email as phishing or legitimate in terms of the following:
    a. Accuracy
    b. Precision
    c. Recall

Table 2

Performance of ePhishSense in terms of Accuracy, Precision, and Recall

Number of Emails	True Positive	True Negative	False Positive	False Negative	Accuracy	Precision	Recall
200	92	104	1	3	98.0%	98.9%	96.8

Table 2 provides detailed information of the system's performance in reviewing 200 emails to determine their legitimacy.

The table shows crucial data such as the number of emails evaluated (200), True Positive (92), which were evaluated as phishing emails, and True Negative (104), which were evaluated as legitimate emails, False Positive (1), which means they were deemed phishing when, in fact, they are legitimate, and False Negative (3), which means they identified phishing emails as legitimate.

The system shows an outstanding capacity to differentiate between phishing and authentic emails, with an accuracy score of 98%. The system's precision in accurately identifying authentic emails is highlighted by its Precision score of 98.9%, and its efficacy in detecting phishing attempts is indicated by its Recall score of 96.8%. These ratings affirm the effectiveness of ePhishSense in safeguarding users against Gmail phishing attempts, validating its reliability based on the applied rating system.

Table 3

Performance Rating of the ePhishSense System

Performance	Percentage	Interpretation (Eboña)
Accuracy	98%	Excellent
Precision	98.9%	Excellent
Recall	96.8%	Superior

Table 3 presents the performance of the ePhishSense system with the interpretation from the Eboña study. The accuracy (98%) of the researchers' system is excellent. The said system achieves an Excellent rating in Precision with a 98.8% rating and a Superior rating in Recall with 96.8%. A precise and comprehensive evaluation of the ePhishSense system's performance was ensured by the metrics used in the rating system, which was inspired by Eboña's study 2013 and provided a credible and relevant framework that seamlessly aligns with the system's objectives.

2. Is there a significant difference in the accuracy between the two studies, "Detecting Phishing Emails Using the O(ND) Algorithm" and "ePhishSense: A Browser Extension Detection System in Gmail for Phishing Emails using Decision Tree Algorithm," considering they utilized same email features for phishing detection?
    a. Sender's email address
    b. Domain Key Identified Message (DKIM)
    c. Outgoing links (URL)

Table 4

Accuracy of the ePhishSense System in Detecting Sender's Email Address

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
1000	0	1000	100%

Table 4 shows the accuracy of the ePhishSense system in detecting sender's email address. It consists of four columns namely Correct Deviations, Incorrect Deviations, total number of deviations, and the accuracy. From a total number of 1000 emails, all of them have been identified correctly by the ePhishSense system providing an output of 100% accuracy in detecting sender's email address.

Table 5

Accuracy of the MightyG8 System in Detecting Sender's Email Address

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
940	60	1000	94%

The table above shows the accuracy of the MightyG8 system in detecting sender's email address. From a total number of 1000 emails, 940 sender's email addresses have

been identified correctly by the ePhishSense system and the remaining 60 emails have been incorrectly detected. The system provides an output of 94% accuracy in detecting sender's email address.

Table 6

Accuracy of the ePhishSense System in Detecting Domain Key Identified Message (DKIM)

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
983	17	1000	98.3%

Table 6 shows the accuracy of the ePhishSense system in detecting Domain Key Identified Message (DKIM). It consists of four columns namely Correct Deviations, Incorrect Deviations, total number of deviations, and the accuracy. From a total number of 1000 emails, 983 of them have been identified correctly by the ePhishSense system and the remaining 17 had been incorrectly identified providing an output of 98.3% accuracy in detecting Domain Key Identified Message (DKIM).

Table 7

Accuracy of the MightyG8 System in Detecting Domain Key Identified Message (DKIM)

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
940	60	1000	94%

The table above shows the accuracy of the MightyG8 system in detecting Domain Key Identified Message (DKIM). From a total number of 1000 emails, 940 sender's email

addresses have been identified correctly by the ePhishSense system and the remaining 60 emails have been incorrectly detected. The system provides an output of 94% accuracy in detecting Domain Key Identified Message (DKIM).

Table 8

Accuracy of the ePhishSense System in Detecting Outgoing Links (URL)

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
956	44	1000	95.6%

Table 8 shows the accuracy of the ePhishSense system in detecting Outgoing links (URL). From a total number of 1000 emails, 956 of them have been identified correctly by the ePhishSense system and the remaining 44 have been incorrectly identified. The system provides an output of 97.5% accuracy in detecting Outgoing links (URL).

Table 9

Accuracy of the MightyG8 System in Detecting Outgoing links (URL)

Correct Deviations	Incorrect Deviations	Total Number of Deviations	Accuracy
918	82	1000	91.8%

The table above shows the accuracy of the MightyG8 system in detecting Outgoing links (URL). From a total number of 1000 emails, 918 sender's email addresses have been identified correctly by the ePhishSense system and the remaining 82 emails have been incorrectly detected. The system provides an output of 91.8% accuracy in detecting Outgoing links (URL).

Table 10

Confusion Matrix for Determining the Accuracy of the ePhishSense System in Detecting Phishing Emails

Testing	Evaluation of System				Accuracy
	True Positive	True Negative	False Positive	False Negative	
200	92	104	1	3	98%

The ePhishSense was tested using 200 emails that consist of both phishing and legitimate emails. 92 emails were classified as True Positive, which were evaluated as phishing emails. 104 emails were classified as True Negative, which was evaluated as legitimate emails. One email was classified as False Positive, which was evaluated as a phishing email but is legitimate in reality. Three emails were classified as False Negative, which was evaluated as legitimate when in fact they are phishing emails.

Table 11

Confusion Matrix for Determining the Accuracy of the MightyG8 in Detecting Phishing Emails

Testing	Evaluation of System				Accuracy
	True Positive	True Negative	False Positive	False Negative	
200	36	152	12	0	94%

The MightyG8 was tested using 200 emails, all of which were both phishing and legitimate. 36 emails were identified as True Positives, signifying that they were deemed to be valid. A total of 152 emails were categorized as True Negatives, signifying their potential malevolent nature. Twelve emails were categorized as False Positives, meaning

they were deemed secure and legitimate when, in reality, they contained malicious code. With a 94% accuracy in detecting emails, the accuracy was calculated utilizing the collected data.

The accuracy of ePhishSense is 98% and the accuracy of MightyG8 is 94%. The difference was then computed using T-Test with a significance of 0.05.

Figure 5. T-test Result

T-Test

Group Statistics

	software	N	Mean	Std. Deviation	Std. Error Mean
evaluation	ePhishSense	200	.98	.140	.010
	MGT8	200	.94	.247	.017

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
evaluation	Equal variances assumed	21.065	.000	2.239	398	.026	.045	.020	.005	.085
	Equal variances not assumed			2.239	315.263	.026	.045	.020	.005	.085

Figure 5 shows two tables namely Group Statistics and Independent Sample Test. For the first table, it displays that there is an equal sample size of 200 emails for both the systems. ePhishSense shows an output of 0.98 mean and 0.140 standard deviation, while MightyG8 has an output of 0.94 mean and 0.247 standard deviation. This indicates that ePhishSense has a more consistent outcome than MightyG8. In the second table, the t-test evaluation comparing ePhishSense to MightyG8 outputs a p-value of 0.026, indicating that ePhishSense has a significant difference from MightyG8.

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presented the summary of findings of the gathered data after the experiment. Presented here also the conclusion based on the results and the recommendation of this research for future development.

Summary of Findings

The purpose of the study is to develop a browser extension that can identify phishing emails in the Gmail platform and notify the user by utilizing the Decision Tree Algorithm. After a thorough evaluation and analysis of the results of this research, following findings are as follows:

1. The results of the performance of the ePhishSense.

   With a total of 200 emails evaluated, there were 92 True Positives, 104 True Negatives, 1 False Positive, and 3 False Negatives. With these, the system exhibited a remarkable accuracy of 98% which indicates that the extension successfully differentiated between legitimate and phishing emails. Then, a precision of 98.9%, a high precision rate signifies the system's efficacy in reducing false positives, and a recall of 96% which indicates the system's ability to accurately identify phishing attempts while limiting the occurrence of false negatives.

2. Significant Difference between the ePhishSense and MightyG8.

   There is a notable difference between ePhishSense and MightyG8 regarding their accuracy in terms of: sender’s email address (100% and 94%, respectively), domain key identified message (98.3% and 94%), and

outgoing links (95.6% and 91.8%). Additionally, based on statement problem number 2, it answers whether there is a significant difference in accuracy between these two. With the use of a larger dataset, ePhishSense obtained a 98% accuracy rate, surpassing MightyG8, which achieved a 94% accuracy rate, with this, there is a 4% difference in accuracy between ePhishSense and MightyG8. The standard deviation for ePhishSense is 0.140, compared to MightyG8's 0.247. Lower standard deviation indicates a less variability and samples are closer to the mean. The p-value for the ePhishSense is 0.026 which is lower than the provided significance level of 0.05. A lower p-value suggests a significant difference between the systems.

Conclusions

Based on the findings of the study, the researchers have arrived on the following conclusions:

1. The accuracy and precision of the tool are excellent according to the rating system by (Eboña, 2013), and for recall, the interpretation is superior. This indicates that the performance of the system showed a high result in terms of the said performance matrix. Therefore, the ePhishSense is reliable for classifying an email as phishing or legitimate.
2. The difference between the ePhishSense and the MightyG8 are noted as follows:
    a. Both tool’s accuracy are within the positive range based on the rating system (Eboña, 2013). However, there is a specific distinction with the two where the ePhishSense is classified as “excellent” while the MightyG8 is “superior”. Based on the rating system, “excellent” is better than “superior”,

thus, ePhishSense emerges as notably more reliable when compared with the MightyG8.

b. The dataset used on the ePhishSense is larger than the MightyG8 and the resulting accuracy has a higher percentage in favor of ePhishSense. Thus, there is an increase in accuracy if the dataset is larger.

c. The ePhishSense has a lower standard deviation compared with MightyG8. This means that the said tool is more clustered around the mean and shows greater consistency. Thus, picking a random sample from the group will yield to a result that is close to the mean.

d. Lastly, as indicated by the result of the T-test, the resulting p-value is lower than the significant level, indicating a statistically significant difference between the systems. Therefore, the null hypothesis is rejected.

Ultimately, the researchers came up with the notion that ePhishSense is an effective tool for detecting phishing emails that can help those who have poor digital literacy and are vulnerable to phishing, especially those who are millennials and gen-z internet users ranging 18 – 40 years old. In light of these findings, the researchers confidently reject the null hypothesis and affirm the superior efficacy of ePhishSense as a robust solution for identifying phishing threats in email communications

Recommendations

From the summary of findings, the researchers recommend the following for improving the system’s performance:

1. Enhance the system by extending its functionality to include the "All Mail" category, rather than just focusing on the "Inbox," which is already filtered from Promotions, Updates, Socials, and Forums.

2. Enhance the process of gathering Blacklisted URLs by creating an API that supports incremental collection and learning, aiming to improve the database of phishing emails.
3. Implement phishing detection in other email service providers such as Yahoo! and Outlook.
4. Continuous training of dataset in order to enhance the system so it can adapt to new phishing techniques.
5. Optimize the decision tree model by adding more parameters such as Presence of JavaScript, Number of dots in Domain name, Unix Magic Attachment, X-mailer, SPF Record Validation, and more.
6. Provide larger training dataset to produce better system’s accuracy.

To deploy the system more efficiently, acquiring cloud services and Gmail API would eliminate the need for manually inputting the "Subject" when scanning emails. By utilizing cloud resources and Gmail API, the system can automate the process of web scraping, such as subject lines, headers, and content, streamlining the email scanning process. This automation not only saves time but also reduces the risk of errors that may occur during manual data entry. Additionally, cloud services provide scalability, ensuring that the system can handle a growing volume of emails and adapt to changing requirements.

Overall, integrating cloud services into the system's deployment strategy not only improves efficiency but also enhances its reliability and scalability.

REFERENCES

A hybrid firefly and support vector machine - ProQuest. (n.d.). Www.proquest.com. Retrieved June 30, 2023, from https://www.proquest.com/docview/2250597207/fulltextPDF/CD2C8CCEC4CF4CCCPQ/30

Abdulrahman, S. H., & Salim, M. (2022). Using Decision Tree Algorithms in Detecting Spam Emails Written in Malay: A Comparison Study. *ITM Web of Conferences*, *42*, 01001. https://doi.org/10.1051/itmconf/20224201001

Adewumi, O. A., & Akinyelu, A. A. (n.d). A hybrid firefly and support vector machine classifier for phishing email detection. https://www.proquest.com/docview/2250597207/fulltextPDF/CD2C8CCEC4CF4CCCPQ/30?accountid=206735

Ahamed, A., Mallya, R., Shetty, A. A., DSouza, D., & Gopi, A. T. (2022, June). Phishing Detection using Decision Tree Model. *International Research Journal of Engineering and Technology (IRJET)*, *9*(6), 2459. Academia.edu. https://www.academia.edu/87907945/Phishing_Detection_using_Decision_Tree_Model

Al-Gubri, Q., M.A.M. Ariff, & Saeh, I. S. (2016). *Performance analysis of machine learning algorithms for power system dynamic security assessment*. Stevenage: The Institution of Engineering & Technology. doi:https://doi.org/10.1049/cp.2016.1294

Allied Telesis. (2018). URL Feature Overview Guide. Retrieved from https://www.alliedtelesis.com/sites/default/files/documents/configuration-guides/url_feature_overview_guide_revc.pdf

Amin, R. M. (2010). *Detecting targeted malicious email through supervised classification of persistent threat and recipient oriented features* (Order No. 3428188). Available from ProQuest Central. (817552431). Retrieved from https://www.proquest.com/dissertations-theses/detecting-targeted-malicious-email-through/docview/817552431/se-2

Bhadane A, Mane SB. State of research on phishing and recent trends of attacks. *i-Manager's Journal on Computer Science.* 2017;5(4):14–35.

Bhavsar, Vaishnavi, Aditya Kadlak, and Shabnam Sharma. "Study on phishing attacks." *Int. J. Comput. Appl* 182 (2018): 27-29.

Boyle, P., & Shepherd, L. A. (2021). MailTrout: A Machine Learning Browser Extension for Detecting Phishing Emails. *Electronic Workshops in Computing*. https://doi.org/10.14236/ewic/hci2021.10

Calvadores & De Vera (2018). Detecting Phishing Emails Using O(ND) Difference Algorithm.

Chakraborty, S., & Mondal, B. (2012). Spam mail filtering technique using different decision tree classifiers through data mining approach-a comparative performance analysis. *International Journal of Computer Applications*, *47*(16).

CISA INSIGHTS Enhance Email & Web Security AT-A-GLANCE RECOMMENDATIONS Adopt a Minimum DMARC Policy of "p=none" Implement HTTPS With HSTS Across All External-Facing Domains Disable Weak Encryption Standards for Web and Email Maintain Ongoing Visibility of DMARC Findings and Reports CYBERSECURITY THREAT. (n.d.). https://www.cisa.gov/sites/default/files/publications/CISAInsights-Cyber-EnhanceEmailandWebSecurity_S508C-a.pdf

Emily Finlay (2021, June 24). Your introductory guide to the waterfall methodology | MindManager. https://blog.mindmanager.com/waterfall-methodology/

Griffin, Charles. "The Latest Phishing Statistics (Updated January 2023) | AAG IT Support." *Aag-It.com*, 6 Feb. 2023, aag-it.com/the-latest-phishing-statistics/#:~:text=With%20an%20average%20of%20%24136.

Chaudhry, J. A., Chaudhry, S. A., & Rittenhouse, R. G. (2016). Phishing attacks and defenses. *International journal of security and its applications*, *10*(1), 247-256.

Chen, H., & Hossain, M. (2021). Developing a Google Chrome Extension for Detecting Phishing Emails. Background and Why Chrome Extension. Retrieved from https://easychair.org/publications/download/RJLM

Darabi, M. M., Izadi, M., & Shahrivari, V. (2020). Phishing detection using machine learning techniques. arXiv preprint arXiv:2009.11116.

Dhulekar, P., Gandhe, S. T., & Mahajan, U. P. (2018). *Development of bottle recycling machine using machine learning algorithm*. Piscataway: The Institute of Electrical and Electronics Engineers,Inc.(IEEE).Retrievedfrom https://www.proquest.com/conference-papers-proceedings/development-bottle-recycling-machine-using/docview/2132065559/se-2

Emily Finlay (2021, June 24). Your introductory guide to the waterfall methodology | MindManager. https://blog.mindmanager.com/waterfall-methodology/

E. Karen et al., “Named-entity recognizer (NER) for Filipino novel excerpts using maximum entropy approach,” Journal of Industrial and Intelligent Information, vol. 1, no. 1, March 2013, pp. 63-67

Gurjar, N. S., Sudheendra, S. S. R., & Kumar, C. S. (2021, July). WebSecAsst-A Machine Learning based Chrome Extension. In *2021 6th International Conference on Communication and Electronics Systems (ICCES)* (pp. 1631-1635). IEEE.

Hu, Y., Guo, C., Ngai, E. W., Liu, M., & Chen, S. (2010). A scalable intelligent non-content-based spam-filtering framework. Expert Systems With Applications, 37(12), 8557–8565. https://doi.org/10.1016/j.eswa.2010.05.020

Ismail, S. S. I., Mansour, R. F., Abd El-Aziz, R. M., & Taloba, A. I. (2022). Efficient E-Mail Spam Detection Strategy Using Genetic Decision Tree Processing with NLP Features. *Computational Intelligence and Neuroscience*, *2022*, e7710005. https://doi.org/10.1155/2022/7710005

Javed, M. (2020, October 19). The Best Machine Learning Algorithm for Email Classification. Towards Data Science. Retrieved from https://towardsdatascience.com/the-best-machine-learning-algorithm-for-email-classification-39888e7b1846

Karim, Abdul, et al. “Phishing Detection System through Hybrid Machine Learning Based on URL.” *IEEE Access*, vol. 4, no. CC BY-NC-ND 4.0, 2023, pp. 1–1, https://doi.org/10.1109/access.2023.3252366.

Krishna, B. P. G., Rajesh, D. S., & Kumar, K. P. (2021, July). Phishing Detection: Machine Learning Implementation. *International Journal of Research in Engineering, Science and Management*, *4*(7), 176. Bielefield Academic Search Engine. https://journal.ijresm.com/index.php/ijresm/article/view/1024/989

Kumar, H., Prasad, A., Rane, N., Tamane, N., & Yeole, A. (2021, September 22). Dr. Phish: Phishing Website Detector. *The 4th International Conference of Computer Science and Renewable Energies*, *297*, 6-9. https://doi.org/10.1051/e3sconf/202129701032

Kutlucan, G., Kaya, O. C., Ilyas, O., Can, K. A., Onursal, C., & Ismail, K. (2023). Species-level microfossil prediction for globotruncana genus using machine learning models. *The Arabian Journal for Science and Engineering.Section B, Engineering, 48*(2), 1315-1332. doi:https://doi.org/10.1007/s13369-022-06822-5

Li, L., Zheng, Y., Li, C., Lu, H., Li, S., & Zhang, L. (2022). A deep ensemble learning-based automated malicious email detection framework. Journal of Network and Computer Applications, 196, 107022. https://doi.org/10.1016/j.jnca.2022.107022

M. Almukaynizi, M. Tariq, S. Alkhalifa and A. Al-Salman (2022). "Detection of Spear Phishing Emails Using a Hybrid Deep Learning Classifier," arXiv preprint arXiv:2201.10752.

Mahajan, R., & Siddavatam, I. (2018). Phishing website detection using machine learning algorithms. *International Journal of Computer Applications*, *181*(23), 45-47.

Maurya, S. (2021). Browser Extension based Hybrid Anti-Phishing Framework using Feature Selection. Ggsipu. https://www.academia.edu/51134471/Browser_Extension_based_Hybrid_Anti_Phishing_Framework_using_Feature_Selection

Maurya, S., Saini, H. S., & Jain, A. (2019). Browser Extension based Hybrid Anti-Phishing Framework using Feature Selection. *(IJACSA) International Journal of Advanced Computer Science and Applications*, *10*(11).

Most Popular Web Browsers in 2021 [July '21 Update] | *Oberlo*. (n.d.). Www.oberlo.com. https://www.oberlo.com/statistics/browser-market-share

Musa, F. B., Abdulkadir, S. A., Muhammed, S., & Musa, A. B. (2022). Anomaly Detection in Emails Using Multi and One-Class Learning Approaches. arXiv preprint arXiv:2203.10408.

N. Feng, "Human Resource Intelligent Scheduling Algorithm Based on Decision Tree Algorithm," *2022 IEEE 2nd International Conference on Mobile Networks and Wireless Communications (ICMNWC)*, Tumkur, Karnataka, India, 2022, pp. 1-4, doi: 10.1109/ICMNWC56175.2022.10031690.

O.K. Sahingoz, E. Buber, O. Demir, and B. Diri (2019). Machine learning-based phishing detection from URLs. Expert Systems with Applications, 117, 345–357. https://doi.org/10.1016/j.eswa.2018.09.029

Ollmann, G. (2004). The phishing guide understanding & preventing phishing attacks abstract. USA. Available at: http://www.ngsconsulting.com.

Pavan, M., Venkata Sai Mounika, B., & Mahitha, G. (2023). Chrome Extension for Detecting Phishing Websites. Journal of Computer Science Engineering, 12(03), 212. Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4398136

Rose, M. A. S. R., Basir, N., Heng, N. F. N. R., Zaizi, N. J. M., & Saudi, M. M. (2022, June). Phishing Detection and Prevention using Chrome Extension. In *2022 10th International Symposium on Digital Forensics and Security (ISDFS)* (pp. 1-6). IEEE.

Shah, B., Dharamshi, K., Patel, M., & Gaikwad, V. (2020). Chrome Extension for Detecting Phishing Websites. International Research Journal of Engineering and Technology (IRJET), 07(03), 2958. Retrieved from https://www.irjet.net/archives/V7/i3/IRJET-V7I3590.pdf

Shankar, A., Shetty, R., & Nath, B. (2019). A review on phishing attacks. *International Journal of Applied Engineering Research*, *14*(9), 2171-2175.

Unnithan, N. A., Harikrishnan, N. B., Vinayakumar, R., Soman, K. P., & Sundarakrishna, S. (2018, March). Detecting phishing E-mail using machine learning techniques. In *Proc. 1st anti-phishing shared task pilot 4th acm iwspa co-located 8th acm conf. data appl. secur. privacy (codaspy)* (pp. 51-54).

Vibration classification of power plant equipment using decision tree algorithm approach (2021). . Piscataway: The Institute of Electrical and Electronics Engineers, Inc. (IEEE). doi:https://doi.org/10.1109/ICT-PEP53949.2021.9601062

Yang, Z., Qiao, C., Kan, W., & Qiu, J. (2019). Phishing Email Detection Based on Hybrid Features. IOP Conference Series, 252, 042051. https://doi.org/10.1088/1755-1315/252/4/042051

Yuan, Z., Yuan, Q., & Wu, J. (2020, November 12). Phishing Detection on Ethereum via Learning Representation of Transaction Subgraphs. In Home Blockchain and Trustworthy Systems Conference. https://link.springer.com/chapter/10.1007/978-981-15-9213-3_14

APPENDICES

Appendix 1. Research Instrument

Sample Experiment Paper

Experiment Paper #1 Test # ____

Objectives

To measure the performance of the system in detecting phishing emails in terms of Accuracy, Precision, and Recall.

Materials

- Desktop/Laptop
- Web Browser
- Email Inputs
- Paper
- Pen
- Bot Generator
- Github Email Dataset
- ePhishSense Browser Extension
- Decision Tree Program
- (2) Test Email Accounts.

Input	True Positive (TP)	True Negative (TN)	False Positive (FP)	False Negative (FN)	Accuracy	Precision	Recall
Emails							

Procedure:

1. Gather the email dataset by executing the bot generator tool and collecting emails from the provided Github dataset.
2. Make a csv file consisting of four columns namely: Email Data, Sender Address, DKIM, URL, and Label.
3. Annotate the emails per column if the said values are existing or not.

4. Execute the Decision Tree program to output a Decision Tree Model with the said csv file as the input.
5. Classify the email based on the displayed output based on the following:
    - **True Positive (TP):** System correctly identifies a phishing email as phishing.
    - **True Negative (TN):** System correctly identifies a legitimate email as legitimate.
    - **False Positive (FP):** System incorrectly identifies a legitimate email as phishing.
    - **False Negative (FN):** System incorrectly identifies a phishing email as legitimate.
6. The software will provide the necessary evaluation sheet in csv format.
7. After classifying all the emails, compute for the Accuracy, Precision, and Recall. The following are the formula to be used:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

Experiment Paper #2 Test # ____

Objectives: To measure the accuracy of the tool in detecting phishing emails compared with Calvadores & De Vera's Detection System,

Materials:

- PC/Laptop
- Web Browser
- Email inputs
- Paper/Pen
- SPSS Application.

Experiment Data	ePhishSense	MightyG8
Mean		
Standard Deviation		
Significance Level		
T-Value		
Degrees of Freedom		
p-Value		

Procedure:

1. Open the SPSS application.
2. Click the "Variable View" button on the bottom pane. Then, create two variables named "System" and "Match Value". On the System's row, add the values of 0 for ePhishSense and 1 for MightyG8. Change the value of the Measure column for both rows as "Nominal."
3. Go to the "Data View". On the System column input "0" for the 1st up to the 200th row and input "1" for the 201st up to the 400th row.
4. Input the testing data of ePhishSense dataset from the 1st up to the 200th row. Similarly input the testing data of MightyG8 dataset from the 201st up to the 400th row.
5. After inputting all the data needed, click the "Analyze" button. Next, click the "Compare

Means". Then, click "Independent Sample T Test"

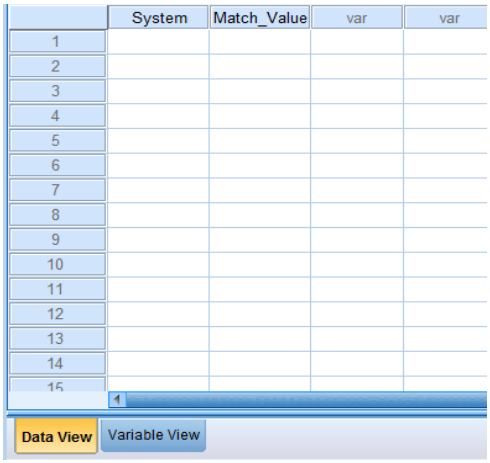
6. An output will be generated. Record the values using the table above.
7. Repeat as necessary.

Dataset Annotation Form

Legend:	**0 - Does not exist/**				
	1 - Exist/Legitimate URLs				
	-1 - Blacklist URLs				
eg.	F1-13, Ethical Clearance	1	1	0	0
ID	**Email Data**	**Sender**	**DKIM**	**URL**	**Label**
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					

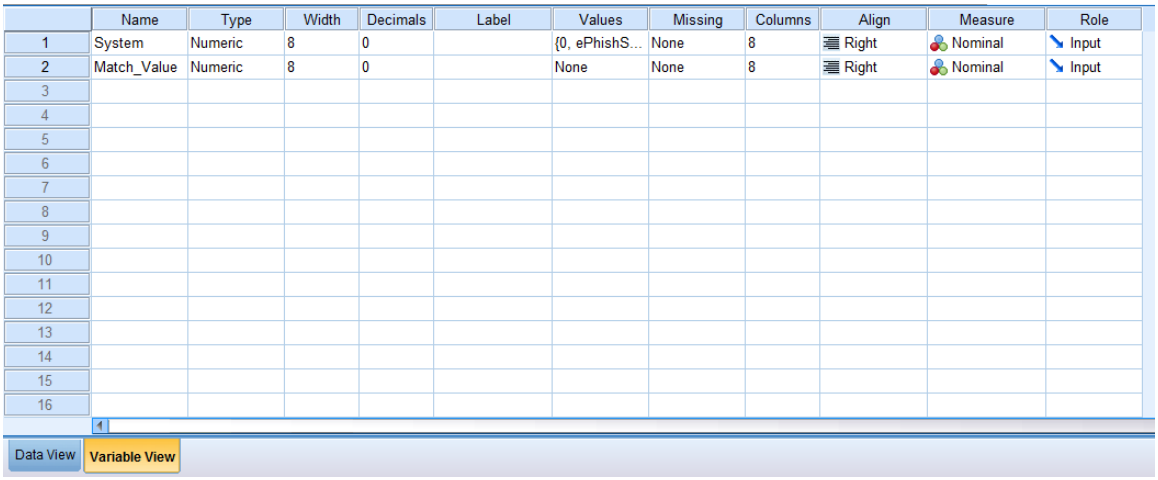
Phishing Emails Evaluation Sheet

ID	Email Subject	Actual Value	Predicted Value	Evaluation	Match Value
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					



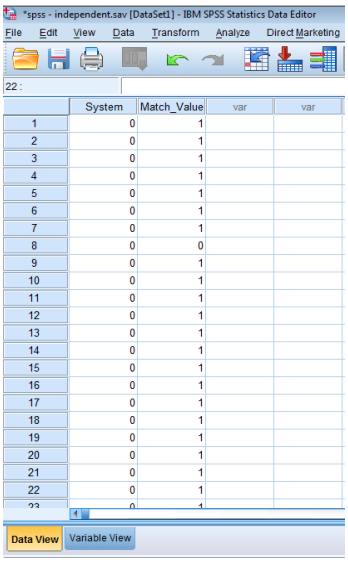
	System	Match_Value	var	var
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Data View | Variable View



	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	System	Numeric	8	0		{0, ePhishS...	None	8	Right	Nominal	Input
2	Match_Value	Numeric	8	0		None	None	8	Right	Nominal	Input
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Data View | Variable View



*spss - independent.sav [DataSet1] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Direct Marketing

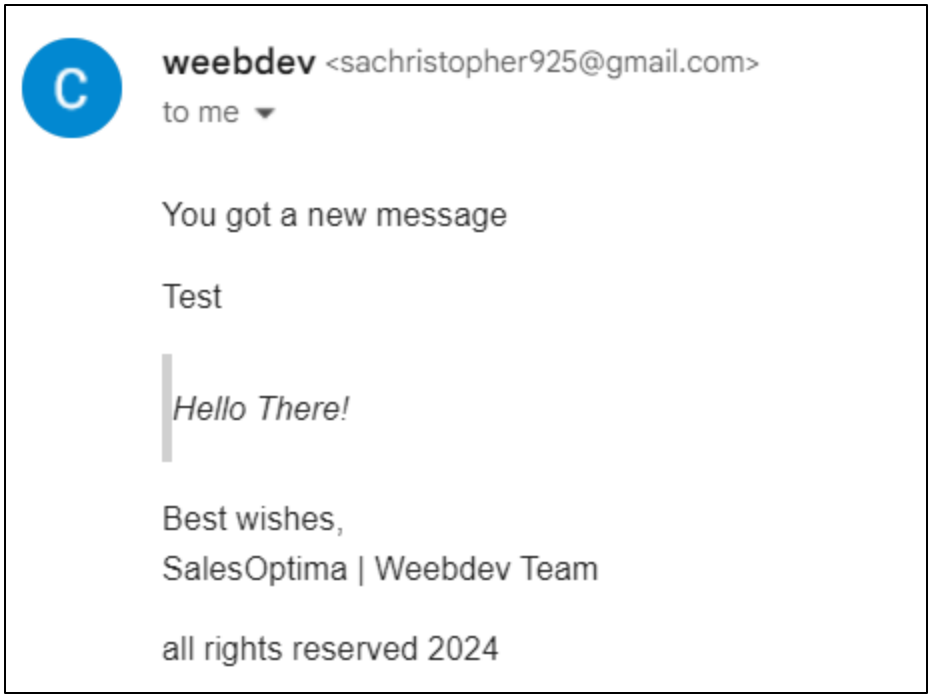
22 :

	System	Match_Value	var	var
1	0	1		
2	0	1		
3	0	1		
4	0	1		
5	0	1		
6	0	1		
7	0	1		
8	0	0		
9	0	1		
10	0	1		
11	0	1		
12	0	1		
13	0	1		
14	0	1		
15	0	1		
16	0	1		
17	0	1		
18	0	1		
19	0	1		
20	0	1		
21	0	1		
22	0	1		

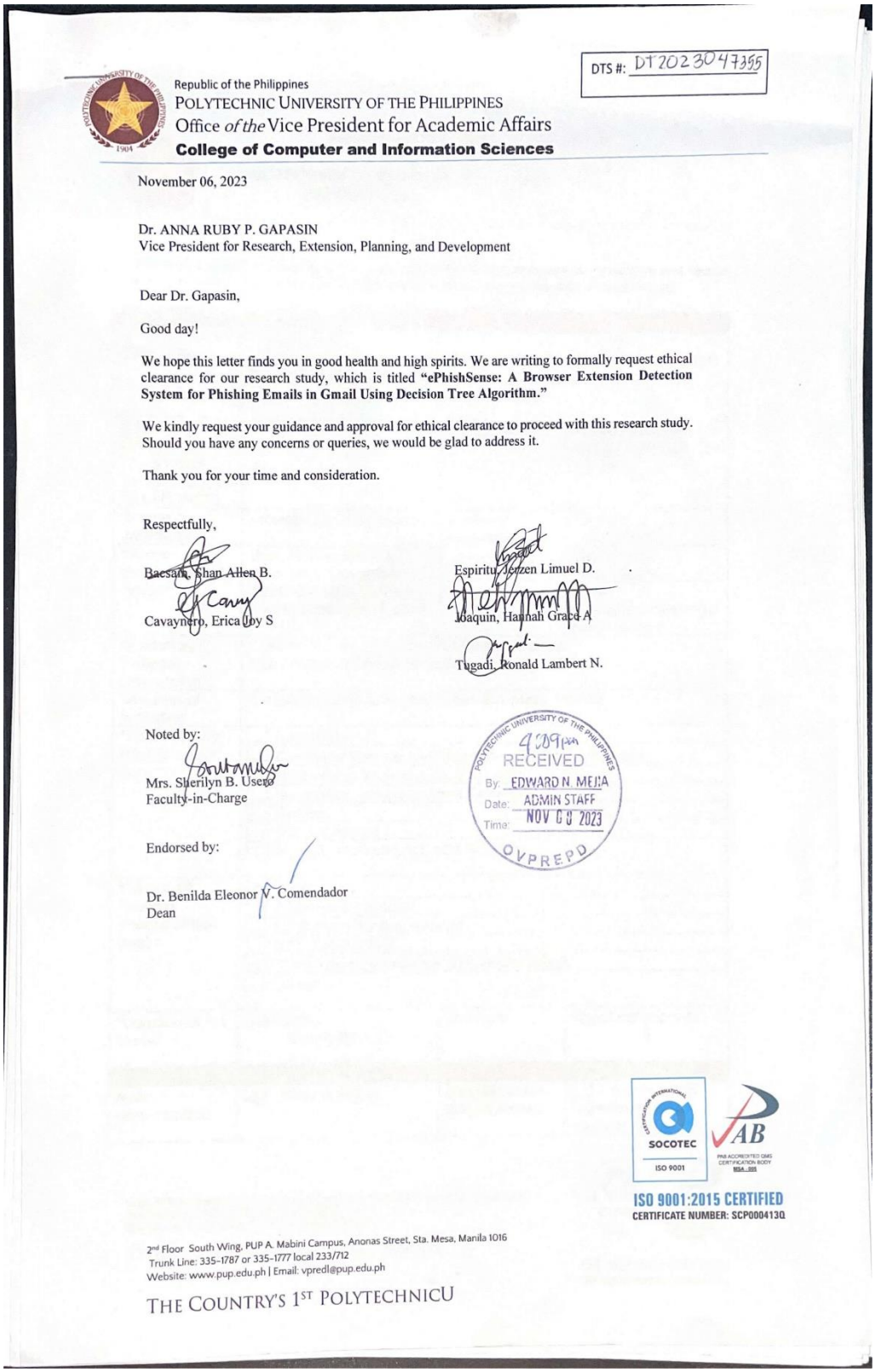
Data View | Variable View

Appendix 2: Raw Data

```
From nobody Sun Jan 25 13:48:11 2015
Return-Path: <root@home.myoswego.com>
X-Original-To: nobody@login.example.com
Delivered-To: nobody@login.example.com
Received: from mail1.example.com (mail1.example.com [10.1.2.3])
 by naughty.example.com (Postfix) with ESMTP id 036B8536EE7
 for <nobody@login.example.com>; Fri, 18 Nov 2005 16:33:46 -0500 (EST)
Received: from mail1.example.com (localhost.example.com [127.0.0.1])
 by mail1.example.com (Postfix) with ESMTP id 35DC06CCF70
 for <nobody@example.com>; Fri, 18 Nov 2005 16:33:43 -0500 (EST)
Received: from home.myoswego.com (cpe-69-202-126-108.twcny.res.rr.com
 [69.202.126.108]) by mail1.example.com (Postfix) with ESMTP
 for <nobody@example.com>; Fri, 18 Nov 2005 16:33:43 -0500 (EST)
Received: by home.myoswego.com (Postfix, from userid 0)
 id D07448F86A; Fri, 18 Nov 2005 16:07:59 -0500 (EST)
From: service@paypal.com
Reply-To: service@paypal.com
MIME-Version: 1.0 Content-Type: text/html\r\n
Content-Type: text/html Content-Transfer-Encoding: 8bit\r\n
Subject: PayPal Account Suspension Notice - PayPal Account Limited
Message-Id: <20051118210759.D07448F86A@home.myoswego.com>
Date: Fri, 18 Nov 2005 16:07:59 -0500 (EST)
To: undisclosed-recipients: ;
X-Spam-Checker-Version: SpamAssassin 3.0.2 (2004-11-16)
X-Spam-Level: ****
X-Spam-Status: No, score=4.1 required=5.0 tests=AWL,HTML_MIME_NO_HTML_TAG,
 MIME_HTML_ONLY,NO_REAL_NAME,RAZOR2_CF_RANGE_51_100,RAZOR2_CHECK,
 RCVD_IN_BL_SPAMCOP_NET,UNDISC_RECIPS autolearn=no version=3.0.2
X-Spam-Report: *  0.0 NO_REAL_NAME From: does not include a real name
```



Appendix 3: Correspondence



DTS #: DT202304739S

Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Office *of the* Vice President for Academic Affairs
College of Computer and Information Sciences

November 06, 2023

Dr. ANNA RUBY P. GAPASIN
Vice President for Research, Extension, Planning, and Development

Dear Dr. Gapasin,

Good day!

We hope this letter finds you in good health and high spirits. We are writing to formally request ethical clearance for our research study, which is titled **"ePhishSense: A Browser Extension Detection System for Phishing Emails in Gmail Using Decision Tree Algorithm."**

We kindly request your guidance and approval for ethical clearance to proceed with this research study. Should you have any concerns or queries, we would be glad to address it.

Thank you for your time and consideration.

Respectfully,

Baesan, Shan Allen B.
Cavaynero, Erica Joy S
Espiritu, Jerzen Limuel D.
Joaquin, Hannah Grace A
Tugadi, Ronald Lambert N.

Noted by:

Mrs. Sherilyn B. Usero
Faculty-in-Charge

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
4:09pm
RECEIVED
By: EDWARD N. MEJIA
Date: ADMIN STAFF
Time: NOV 08 2023
OVPREPD

Endorsed by:

Dr. Benilda Eleonor V. Comendador
Dean

SOCOTEC ISO 9001
PAB ACCREDITED QMS CERTIFICATION BODY
ISO 9001:2015 CERTIFIED
CERTIFICATE NUMBER: SCP000413Q

2nd Floor South Wing, PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line: 335-1787 or 335-1777 local 233/712
Website: www.pup.edu.ph | Email: vpredl@pup.edu.ph

THE COUNTRY'S 1ST POLYTECHNICU

Appendix 4: Ethical Clearance

Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE *of the* VICE PRESIDENT *for* RESEARCH, EXTENSION, *and* DEVELOPMENT
RESEARCH MANAGEMENT OFFICE
UNIVERSITY RESEARCH ETHICS CENTER

Date: December 12, 2023

To/For: Hannah Grace A. Joaquin
Shan Allen B. Bacsain
Erica Joy S. Cavaneyro
Jerzen Limuel D. espiritu
Ronald lambert N. Tugadi

Subject: Ethical Clearance

From: Prof. Jackson Jake U. LLames
Chief, Research Ethics Center

This is to inform you that your submitted documentary requirements for your research project titled **"ePhishSense: A Browser Extension Detection System for Phishing Emails in Gmail Using Decision Tree Algorithm"** passed the evaluation of the PUP Research Ethics Committee (REC) in accordance with the requirements set by the Philippine Health Research Ethics Board (PHREB).

UREC Code	UREC-2023-1216
Type of Review	Expedited
Approval Date	December 12, 2023
Expiry Date	December 11, 2024
PUP-UREC Decision	Approved

The standard conditions of this approval are as follows:

1. Conduct the project strictly in accordance with the submitted and approved research protocol and other documentary requirements.
2. If changes will be done in the conduct of the research project/study that will affect the research participants, an amendment of the research protocol must be submitted to urec@pup.edu.ph before implementing such changes.
3. For ethical clearance that is about to expire, researcher/s must apply for resubmission of the research protocol.
4. A final report/terminal report must be submitted when the research project/study is complete.
5. Researchers must advise in writing the PUP-UREC (email: urec@pup.edu.ph) if the research project/study has been discontinued.

You may now commence on your research project/study. Good luck.

S423, 4th Floor South Wing, PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line: 335-1787 or 335-1777 local 235/357
Website: www.pup.edu.ph | Email: vpredl@pup.edu.ph

THE COUNTRY'S 1st POLYTECHNICU

SOCOTEC ISO 9001
AB
ISO 9001:2015 CERTIFIED
CERTIFICATE NUMBER: SCP0004130

Appendix 5: Certification of Originality

Similarity Report

PAPER NAME

Group 6_ePhishSense: A BROWSER EXT ENSION DETECTION SYSTEM FOR PHIS HING EMAILS IN GMAIL USING DECISIO N TREE ALGORITHM.pdf

WORD COUNT	CHARACTER COUNT
11608 Words	65022 Characters
PAGE COUNT	FILE SIZE
55 Pages	410.5KB
SUBMISSION DATE	REPORT DATE
Jun 10, 2024 5:47 PM GMT+8	Jun 10, 2024 5:48 PM GMT+8

18% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

- 9% Internet database
- 6% Publications database
- Crossref database
- Crossref Posted Content database
- 13% Submitted Works database

Appendix 6: Certification of Editing

CERTIFICATION OF EDITING

This is to certify that the thesis work entitled **ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAILS IN GMAIL USING DECISION TREE ALGORITHM** by **Bacsain, Shan Allen B., Cavaneyro, Erica Joy S., Espiritu, Jerzen Limuel D., Joaquin, Hannah Grace A., Tugadi, Ronald Lambert N.** was proofread and edited by the undersigned.

This certification is being issued for whatever legal purpose it may serve.

Signed: Clay Jan Dela Cruz License No. 2090932

Affiliation: Systems Plus Computer Foundation Inc. - Cubao

Contact Number: 09672942418

Email Address: kleighdelacruz@gmail.com

Date: June 12, 2024

PUP Data Privacy Act

Thank you for providing your data. At Polytechnic University of the Philippines (PUP) we respect and value your rights as a data subject under the Data Privacy Act (DPA). PUP is committed to protecting the personal data you provide in accordance with the requirements under the DPA and IRR. In this regard, PUP implements reasonable and appropriate security measures to maintain the confidentiality, integrity and availability of your personal data. For more detailed Privacy Notice you may visit the following link: https://www.pup.edu.ph/privacy.

Appendix 7: Biographical Statements



BACSAIN. SHAN ALLEN B.

A fourth-year computer science student at the Polytechnic University of the Philippines (PUP). He specializes in backend programming and has technical competency in Python, Java, computer networking, and software development. His experience includes R, Python, and SQL, with a particular emphasis on data visualization and analysis. His academic interests are around dissertation projects and corporate datasets, which aim to facilitate successful decision-making through data-driven insights.



CAVANEYRO, ERICA JOY S.

A Computer Science student with a focus on frontend development, UI and UX design from the Polytechnic University of the Philippines (PUP). Has fundamental backend development Skills. Proficient in programming languages such as C, Java, and Python, with expertise of the React framework. Her scholarly interests include data science and analytics, artificial intelligence, machine learning, software and web development, and human-computer interface.



ESPIRITU, JERZEN LIMUEL D.

A graduating Computer Science student from the Polytechnic University of the Philippines (PUP) with a focus on backend development and skills in Python, Java, and Computer Networking. His personal ability is known for his collaborative collaboration and adaptability. His scholarly interests are on cloud computing and cybersecurity.



JOAQUIN, HANNAH GRACE A.

A graduating Computer Science student at the Polytechnic University of the Philippines (PUP) with expertise in UI design, front-end development, and basic backend development. She is proficient in Java and Python and regularly participates in projects requiring strong analytical and programming abilities. Her scholarly interests in Computer Science include web development and data analytics.



TUGADI, RONALD LAMBERT N.

A fourth-year computer science student at the Polytechnic University of the Philippines (PUP). He is technically proficient in Basic Computer Networking, Java, Python, HTML, and JavaScript. His personal abilities include Figma UI/UX design and Full Stack MERN website development (MongoDB, ExpressJS, ReactJS, NodeJS). He is also excellent at producing presentations, editing, and graphic design.

Appendix 8. Implementation Report

Introduction

In today's digital world, phishing has become a serious problem for both individuals and companies. In the first half of 2022, there was a noticeable increase in phishing attacks on payment systems, online shopping sites, and local banks in the Philippines. This highlights the need for better online security and awareness among people and businesses.

Phishing attacks through emails have caused trouble globally. Even with strong protection, scammers managed to steal $44.2 million in 2021, averaging $136 for each attack. Surprisingly, younger generations like Millennials and Gen-Z are more at risk.

To help people stay safe, there are browser extensions that quickly warn users about suspicious websites and emails. These extensions are especially helpful for users of popular web browsers like Chrome.

Using machine learning algorithms is crucial for online safety. One important method is the decision tree, which helps identify patterns in phishing attacks. By understanding how these attacks work, we can improve our online security.

After seeing the serious problems caused by phishing, researchers want to understand it better. The study used Decision Tree Algorithm to check how well it can detect phishing emails. The researchers' main goal is to create a simple tool, like a browser extension, that can tell if an email is fake. This tool will be carefully tested to make sure it's good at spotting fake emails and helping people stay safe online.

Problem Statement

The goal of the study is to propose a system that detects general emails sent in Gmail and determines its usefulness in protecting users from phishing emails. The study aims to answer the following questions:

1. What is the performance of the system in classifying an email as phishing or legitimate in terms of the following:
    a. Accuracy
    b. Precision
    c. Recall

2. Is there a significant difference in the accuracy between the two studies, "Detecting Phishing Emails Using the O(ND) Algorithm" and "ePhishSense: A Browser Extension Detection System in Gmail for Phishing Emails using Decision Tree Algorithm," considering they utilized same email features for phishing detection?
    a. Sender’s email address
    b. Domain Key Identified Message (DKIM)
    c. Outgoing links (URL)

Time Frame

Table 12.

Time Frame of ePhishSense

Activity	Status	Date
Chapter 1 to 3 Documentation	**Done**	April to July
Developing the tool	**Done**	October to December
Finalizing the tool	**Done**	Month of December
Data Gathering	**Done**	Month of December
Testing/training the Tool	**Done**	Month of December
Improving the accuracy of the tool	**Done**	Month of December
Experimentation	**Done**	Month of December
Chapter 4-5 Documentation	**Done**	December to Present

Implementation Procedure

The researchers successfully executed the system implementation by adhering to the following procedural steps.

1. The researchers gathered 800 emails for training and 200 for testing data from Bot Tool and Github.
2. Then, the researchers manually annotated the data from the gathered emails by checking if the parameters existed or not.
3. The annotated dataset has been used to provide the Decision Tree Model.

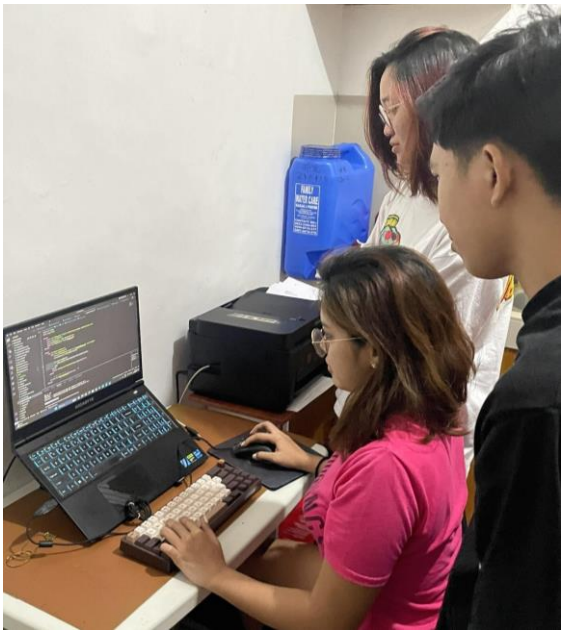
4. An output will be displayed containing all the data that will be needed for the experiment paper.
5. Follow the necessary procedure on the experiment paper and transfer the outputted data from the model.
6. Evaluate the system by comparing the original email label with the label predicted by the system.

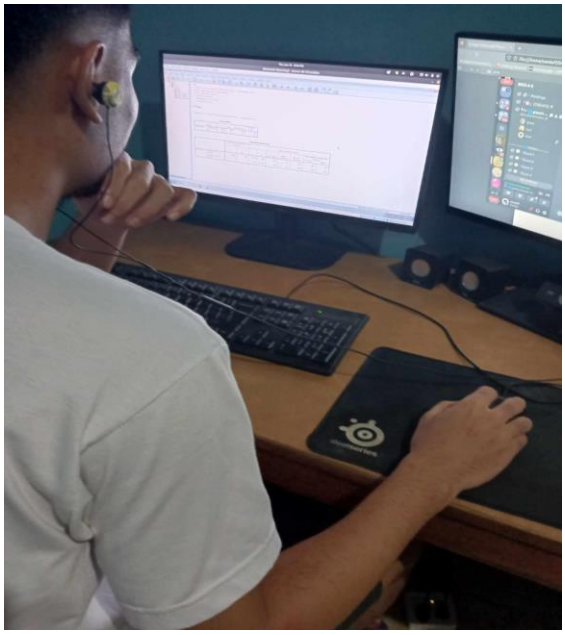
Issues and Concerns

The following are the issues and concerns that were encountered during implementation:

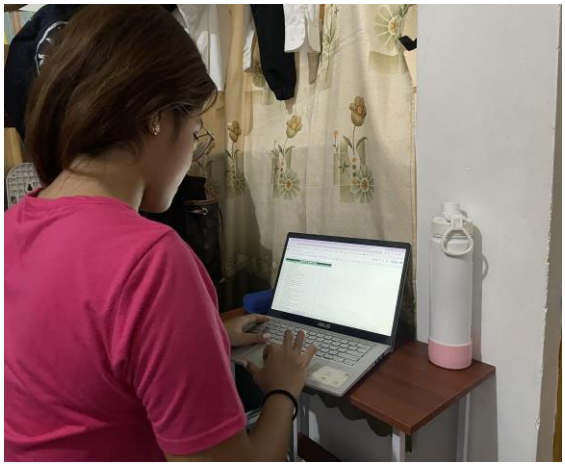
1. Unpacking the software on different browsers resulted in an inconsistent user interface.
2. The bot tool of Calvadores requires an update because it is not working due to not complying with the current Gmail's policies.
3. The first few executions are slightly slower when providing an output.

Proof of Implementation







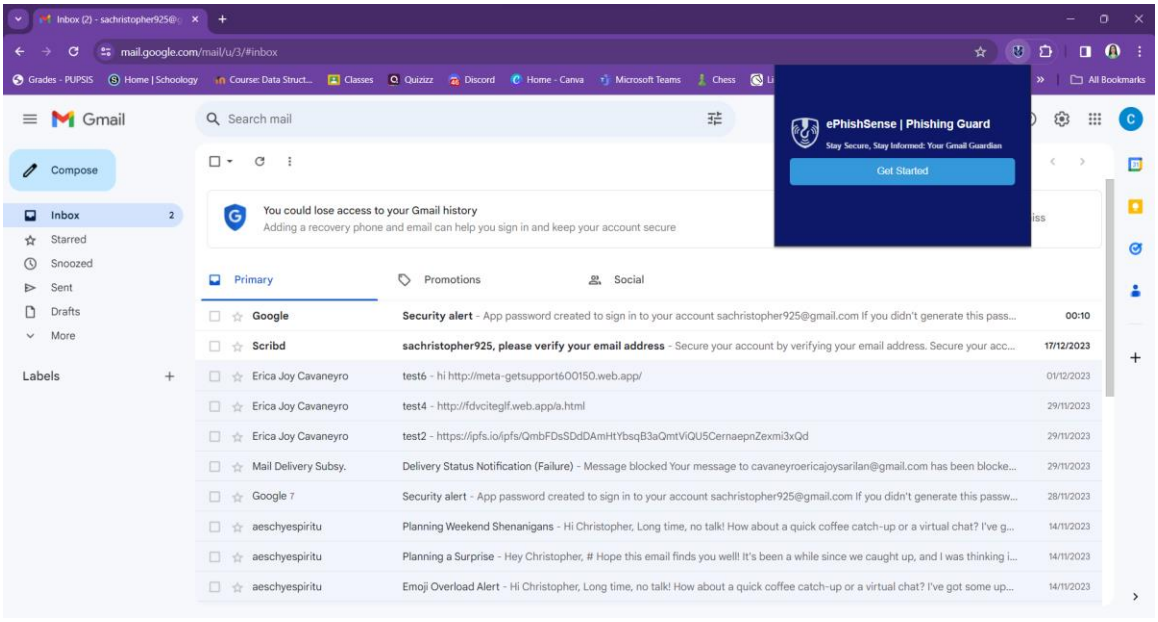


Phishing Emails Evaluation Sheet					
ID	Email Subject	Actual Value	Predicted Value	Evaluation	Match Value
1	F8 - 55. Collect Stars Any Way You Pay	0	0	TN	1
2	F9 -95, phishing0 - 191.eml	0	0	TN	1
3	F5 - Funny Story to Share	0	0	TN	1
4	F6 - phishing2 - 1006.eml	1	1	TP	1
5	F9 -42, phishing0 - 138.eml	1	1	TP	1
6	F5 - 1266. Order Confirmation Alert	1	1	TP	1
7	F12 - 97 - Last call to join us for the free webinar 1	1	1	TP	1
8	F5 - 1130. =?UTF-8?Q?=E2=9A=A0=EF=B8=8F_Urg	1	0	FN	0
9	F8 - 38. Welcome to VYBE!	0	0	TN	1
10	F10-80, PUP Client Satisfaction Survey.eml	0	0	TN	1
11	F6 - phishing2 - 1124.eml	1	1	TP	1
12	F10-37, Erica Joy Cavaneyro _ QA Intern.eml	0	0	TN	1
13	F12 - 68 - New announcement_ _Synchronous Clas	0	0	TN	1
14	F10-5, [ONGOING] Sea the Future Ahead_ The Div	0	0	TN	1
15	F9 -22, phishing0 - 118.eml	1	1	TP	1
16	F7-71, S'well	0	0	TN	1
17	F12 - 26 - Vouchers = savings. ðŸ°*	0	0	TN	1
18	F8 - 70. Get 30% OFF on your next GrabCar ride! U	0	0	TN	1
19	F6 - phishing2 - 1072.eml	1	1	TP	1
20	F5 - For Presentation	0	0	TN	1
21	F5 - 1222. USDT Withdrawal Request Received	1	1	TP	1
22	F6 - phishing2 - 1068.eml	1	1	TP	1
23	F12 - 100 -Connect data now to get the most out o	1	1	TP	1
24	F12 - 89 - Security alert	0	0	TN	1
25	F6 - phishing2 - 1090.eml	1	1	TP	1
26	F7-60, You're my everything and more, my most st	0	0	TN	1
27	F5-1485, =?UTF-8?B?Q2FzZSBOwrA=?=01237: You	1	1	TP	1
28	F2 - Emoji Overload Alert	0	0	TN	1
29	F12 - 12 - Your trial has ended	0	0	TN	1
30	F6 - phishing2 - 1038.eml	1	1	TP	1
31	F5 - 1365. In re FTX Trading Ltd. =?UTF-8?B?4oCT? Reminder	0	0	TN	1
32	F6 - phishing2 - 1014.eml	1	1	TP	1
33	F10-27, Connection Starts Here.eml	0	0	TN	1
34	F5 - 1115. Personalisierte Prognose fÃ¼r schnellen	1	0	FN	0
35	F10-24, Buzzsprout Newsletter_ How Episode Num	0	0	TN	1
36	F10-96, Welcome to Looker Studio.eml	1	1	TP	1
37	F5-1468, Livelo Informa: Voce tem saldo disponive	1	1	TP	1
38	F5-1247, Fw: The 6 Best Ways To Easily Make Mon	1	1	TP	1
39	F10-64,Meracle L Bruce replied to a comment on _	0	0	TN	1
40	F5 - 1381. See What Everyone's Whispering About	0	0	TN	1
41	F9 -90, phishing0 - 186.eml	1	1	TP	1
42	F6 - phishing2 - 1070.eml	1	1	TP	1
43	F6 - phishing2 - 1126.eml	1	1	TP	1
44	F8 - 23. Introducing_ Perplexity Collections	0	0	TN	1
45	F5 - Account Registration	0	1	FP	0
46	F5 - 1245. Fw: The 6 Best Ways To Easily Make Mo $923 million until now)	1	1	TP	1
47	F3 - Nike	0	0	TN	1
48	F9 -39, phishing0 - 135.eml	1	1	TP	1
49	F7-77, Yeti	0	0	TN	1
50	F12 - 72 - New jobs from your November 7, 2023 f	0	0	TN	1
51	F10-36, Due tomorrow _ Online Class Agreement	0	0	TN	1

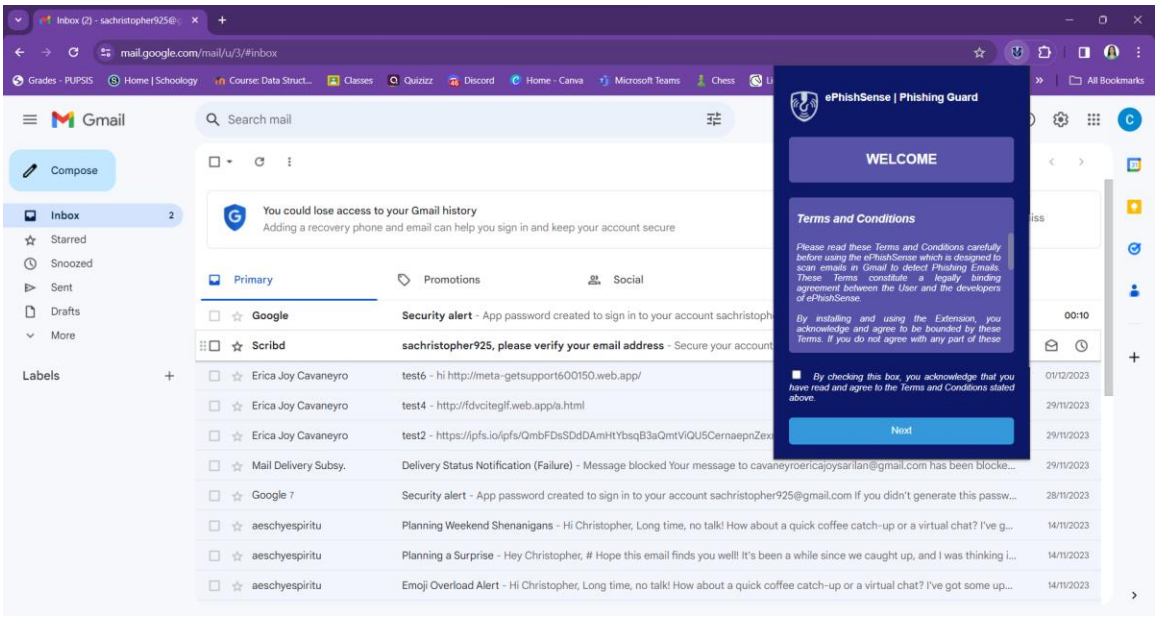
Phishing Emails Evaluation Sheet					
ID	Email Subject	Actual Value	Predicted Value	Evaluation	Match Value
1		1	1	TP	1
2		1	1	TP	1
3		1	1	TP	1
4		1	1	TP	1
5		1	1	TP	1
6		1	1	TP	1
7		1	1	TP	1
8		1	1	TP	1
9		1	1	TP	1
10		1	1	TP	1
11		1	1	TP	1
12		1	1	TP	1
13		1	1	TP	1
14		1	1	TP	1
15		1	1	TP	1
16		1	1	TP	1
17		1	1	TP	1
18		1	1	TP	1
19		1	1	TP	1
20		1	1	TP	1
21		1	1	TP	1
22		1	1	TP	1
23		1	1	TP	1
24		1	1	TP	1
25		1	1	TP	1
26		1	1	TP	1
27		1	1	TP	1
28		1	1	TP	1
29		1	1	TP	1
30		1	1	TP	1
31		1	1	TP	1
32		1	1	TP	1
33		1	1	TP	1
34		1	1	TP	1
35		1	1	TP	1
36		1	1	TP	1
37		0	0	TN	1
38		0	0	TN	1
39		0	0	TN	1
40		0	0	TN	1
41		0	0	TN	1
42		0	0	TN	1
43		0	0	TN	1
44		0	0	TN	1
45		0	0	TN	1
46		0	0	TN	1
47		0	0	TN	1
48		0	0	TN	1
49		0	0	TN	1
50		0	0	TN	1
51		0	0	TN	1
52		0	0	TN	1

Dataset Annotation Form					
Legend:	**0 - Does not exist/Legitimate**				
	1 - Exist/Phishing				
eg.	F1-13, Ethical Clearance	1	1	0	0
ID	**Email Data**	**Sender**	**DKIM**	**URL**	**Label**
1	F7-1, Our exclusive 11.11 celebration starts today!	1	1	1	0
2	F7-2, Introducing: Binance Web3 Wallet	1	1	1	0
3	F7-3, =?UTF-8?B?8J+OgSBQcmltb2dlbXMgaGF2ZSBiZWVuIHNlbr	1	1	1	0
4	F7-4, =?utf-8?Q?=E2=8F=B3Black=20Friday=20Countdown=3A=20	1	1	1	0
5	F7-5, =?UTF-8?B?SW5zaWRlOiBPdXIgYmVzdCBCbGFjayBGcmlk	1	1	1	0
6	F7-6, Discover your next favorite book	1	1	1	0
7	F7-7, Join the One Piece Film Gold Advance Screening!	1	1	1	0
8	F7-8, You can get the resources you want without upgrading	1	1	1	0
9	F7-9, Apply to top growing companies at the Online Career Fair	1	1	1	0
10	F7-10, Escape the everyday with Passenger	1	1	1	0
11	F7-11, Just A Few Easy Steps to Score Your Dream Overseas Job	1	1	1	0
12	F7-12, LAST CHANCE: Get 40% Off Any Individual Plan	1	1	1	0
13	F7-13, Let's get started!	1	1	1	0
14	F7-14, Your PwC AC Manila Journey Starts Here!	1	1	1	0
15	F7-15, Curiousity or curiosity, which one is it?	1	1	1	0
16	F7-16, 2 Days Only: Get 50% Off Academia Premium!	1	1	1	0
17	F7-17, Get Meloetta in December! Find Out How!	1	1	1	0
18	F7-18, [GlassWire Official Forum] Summary	1	1	1	0
19	F7-19, IMDb Tips: Building your Watchlist	1	1	1	0
20	F7-20, This Weekend: Get Lifetime Service for Your PC	1	1	1	0
21	F7-21, =?UTF-8?Q?Doing=20things=20the=20easy=20way?=	1	1	1	0
22	F7-22, Get inspired with these 18 trending ideas	1	1	1	0
23	F7-23, =?UTF-8?Q?Creating=20stellar=20visuals=20just=20got=20	1	1	1	0
24	F7-24, 71% discount - The digital cutter for your images	1	1	1	0
25	F7-25, GeForce Experience PUBG Game Code Giveaway	1	1	1	0
26	F7-26, Cyberlink Powerdirector (PATCH)	1	1	0	0
27	F7-27, Inquiry - CNL Invitation	1	1	0	0
28	F7-28, Email are not case sensitive	1	1	0	0
29	F7-29, Emails are not case sensitive	1	1	0	0
30	F7-30, Exam cancelled	1	1	0	0
31	F7-31, Re: Exam cancelled	1	1	0	0
32	F7-32, =?utf-8?b?TmV3IGxvZ2lu?=	1	1	0	0
33	F7-33, Hey Friends!	1	1	0	0
34	F7-34, Simple test message	1	1	0	0
35	F7-35, Hello from Python	1	1	0	0
36	F7-36, Thank you for your interest in the organization	1	1	0	0
37	F7-37, Simple test message again	1	1	0	0
38	F7-38, Hey Friends, What's Up	1	1	0	0
39	F7-39, How's Life Treating You?	1	1	0	0
40	F7-40, Let's Make Plans!	1	1	0	0
41	F7-41, Funny Story to Share	1	1	0	0
42	F7-42, Planning Weekend Shenanigans	1	1	0	0

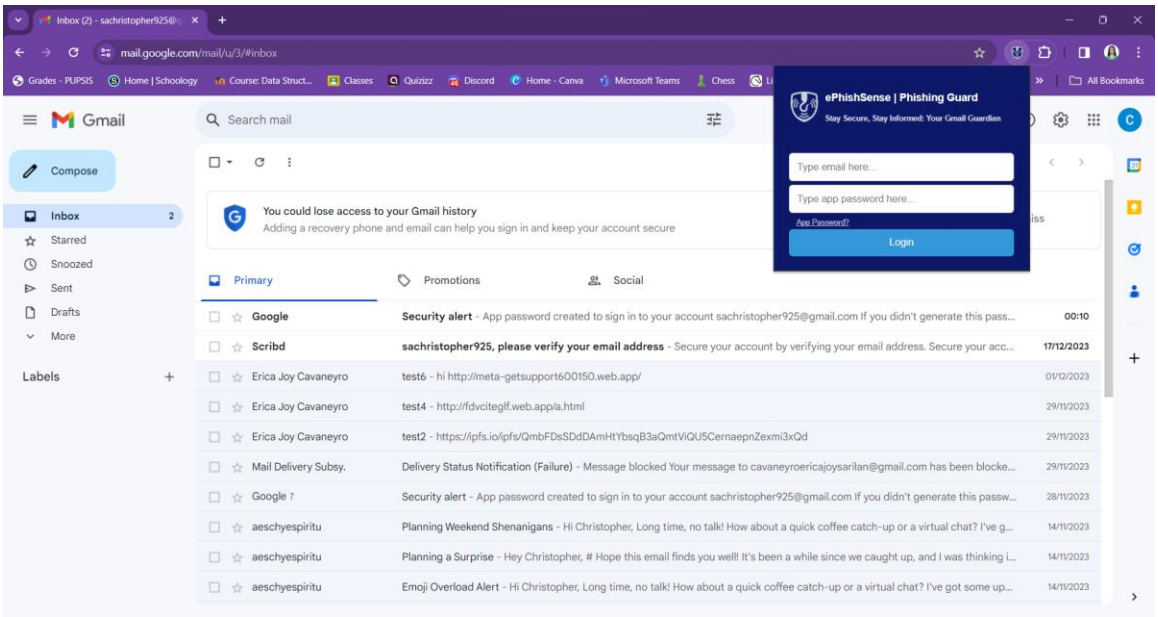
Appendix 9. Screen Layout and Description



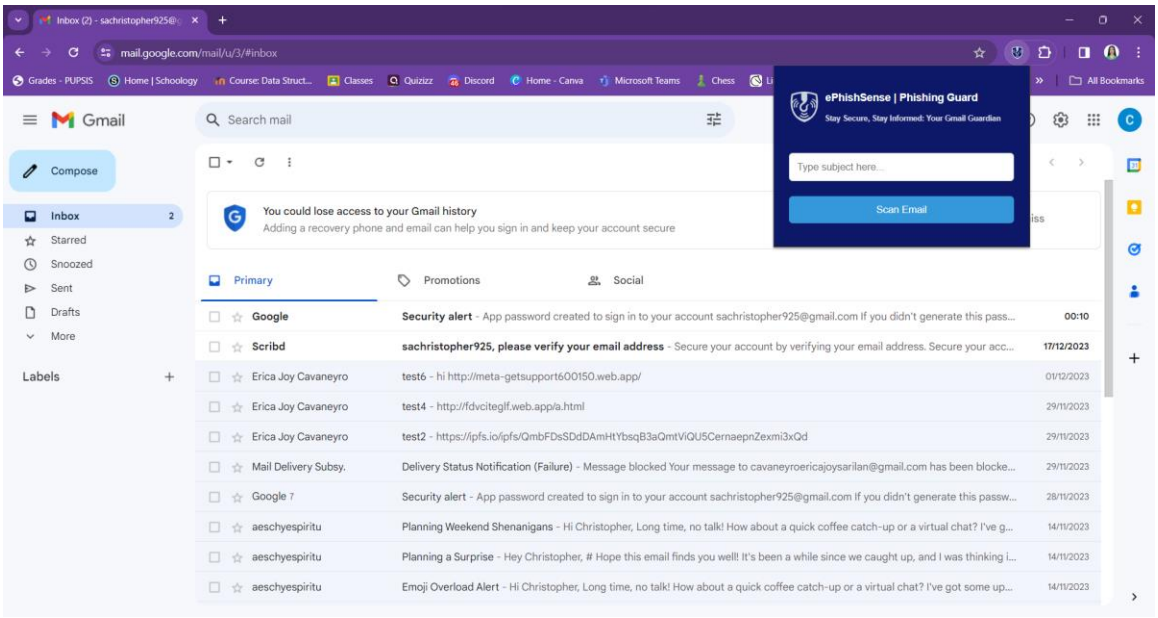
Get Started page. This pop-up page showcases the name of the browser extension and its tagline, along with a clickable button to proceed to the next page.



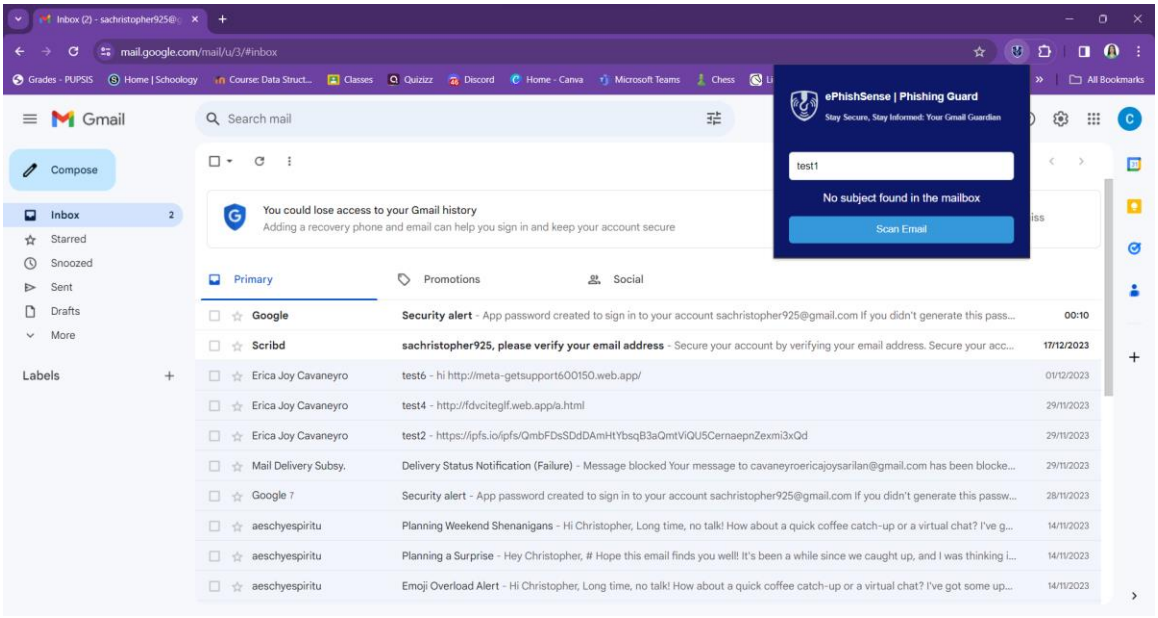
Welcome page. This pop-up page allows users to read the terms and conditions of the browser extension and lets them decide whether they acknowledge and agree to the terms and conditions.



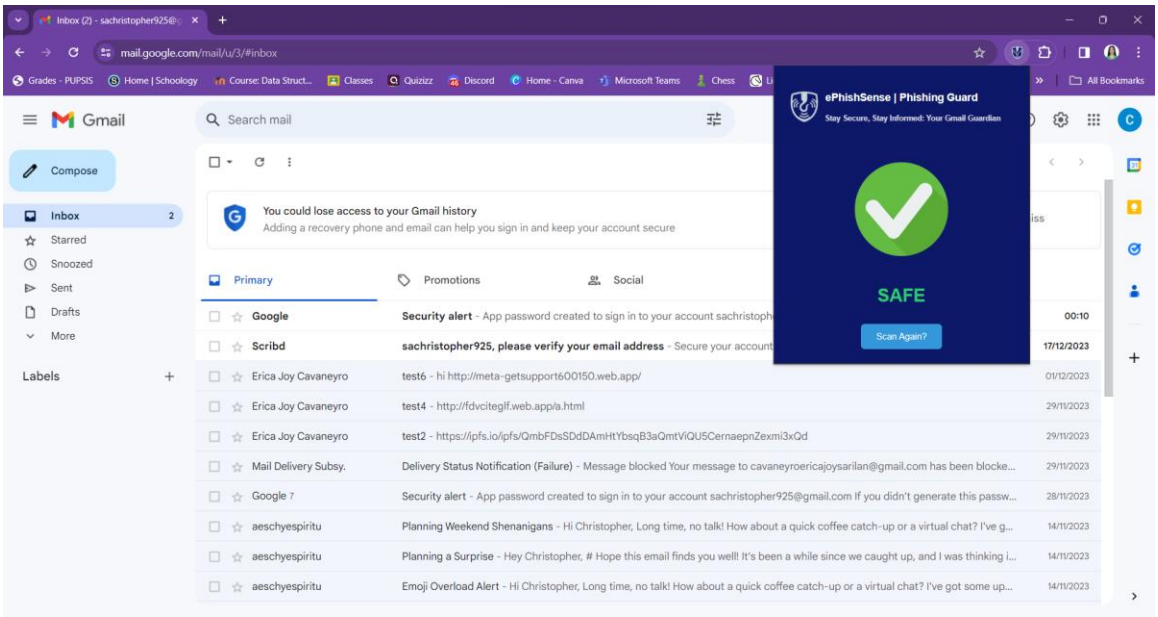
Login Page. This pop-up page allows users to input their credentials, such as email and app password. It also has a clickable link that redirects you on how to get an app password.



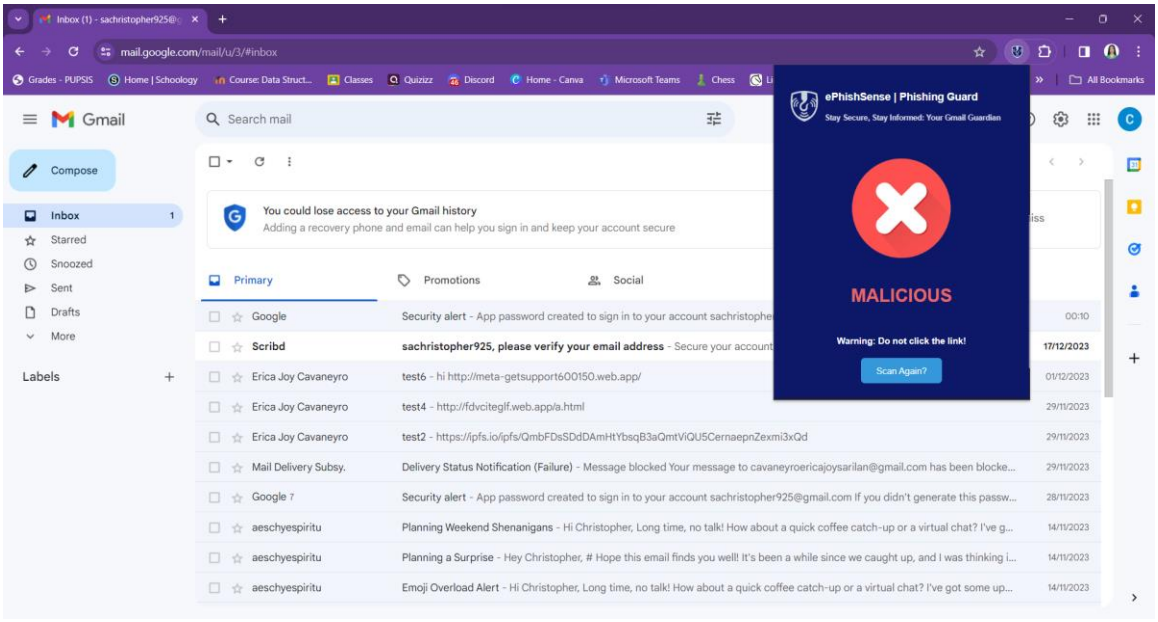
Scan Page. This pop-up page allows users to input the subject of an email that they want to scan.



Scan Page (Error Handling). This pop-up page shows whether the subject is existing or not



Result page. This pop-up page will show "SAFE" when the email is legitimate.



Result page. This pop-up page will show "MALICIOUS" when the email is not safe or is malicious.

Appendix 10. Thesis Revision Matrices

Proposal Defense Revision Matrix

THESIS PROPOSAL DEFENSE REVISION MATRIX

PROPOSAL TITLE				SECTION		GROUP NO.	
ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAILS IN GMAIL USING DECISION TREE ALGORITHM				BSCS 3-2		Group 6	
Document Part	**COMMENTS/SUGGESTIONS/RECOMMENDATIONS by the Panel**	**ACTION TAKEN by the Proponent/s**	**Page No.**	**REMARKS (Panelist Approval by indicating the date)**			
				Panel 1	Panel 2	Panel 3	Panel 4
Title	The system should be available on all browsers	Changed Chrome browser into All Browsers.					
Preliminaries							
Chapter 1	SOP #2 – Check the parameters of the study you are comparing	Use the same elements of email used in the chosen study to be compared to the current study.					
	Ensure that the features included in your study are distinct from those in previous and existing studies	Added other features of an email.	7, 8				
Chapter 2							
Chapter 3	When testing the system, ensure that the email account to be used remains uncompromised	A new email account will be used specifically for testing the phishing detection.	42				
	In the system architecture, thoroughly discuss security measures during system testing	Briefly discussed security measures during system testing.	42				
References							
Appendices							
General Observations							

(Please use additional sheet if necessary)

Legend:

Name of Panelist 1: Prof. Iluminada Vivien R. Domingo

Name of Panelist : Prof. Marian G. Arada

Name of Panelist 3: Prof. John Dustin D. Santos

Name of Panelist 4: Prof. Lydinar D. Dastas

Tool Defense Revision Matrix

Polytechnic University of the Philippines
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
Sta. Mesa, Manila

BSCS THESIS TOOL REVISION MATRIX

PROPOSAL TITLE	SECTION	GROUP NO.
ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAILS IN GMAIL USING DECISION TREE ALGORITHM	4-2	6

Proposal/Tool Part	COMMENTS/SUGGESTIONS/RECOMMENDATIONS by the Panel	ACTION TAKEN by the Proponent/s	Page No.	REMARKS (Panelist Approval by indicating the date)			
				Panel 1	Panel 2	Panel 3	Panel 4
Title*							
Preliminaries							
Chapter 1*							
Chapter 2*							
Chapter 3*							
Tool (Software/Hardware)							
References							
General Observations	Scan other mailbox, not just the "inbox" of Gmail.	Will consider in recommendation					L.DASTAS

(Please use additional sheet if necessary)

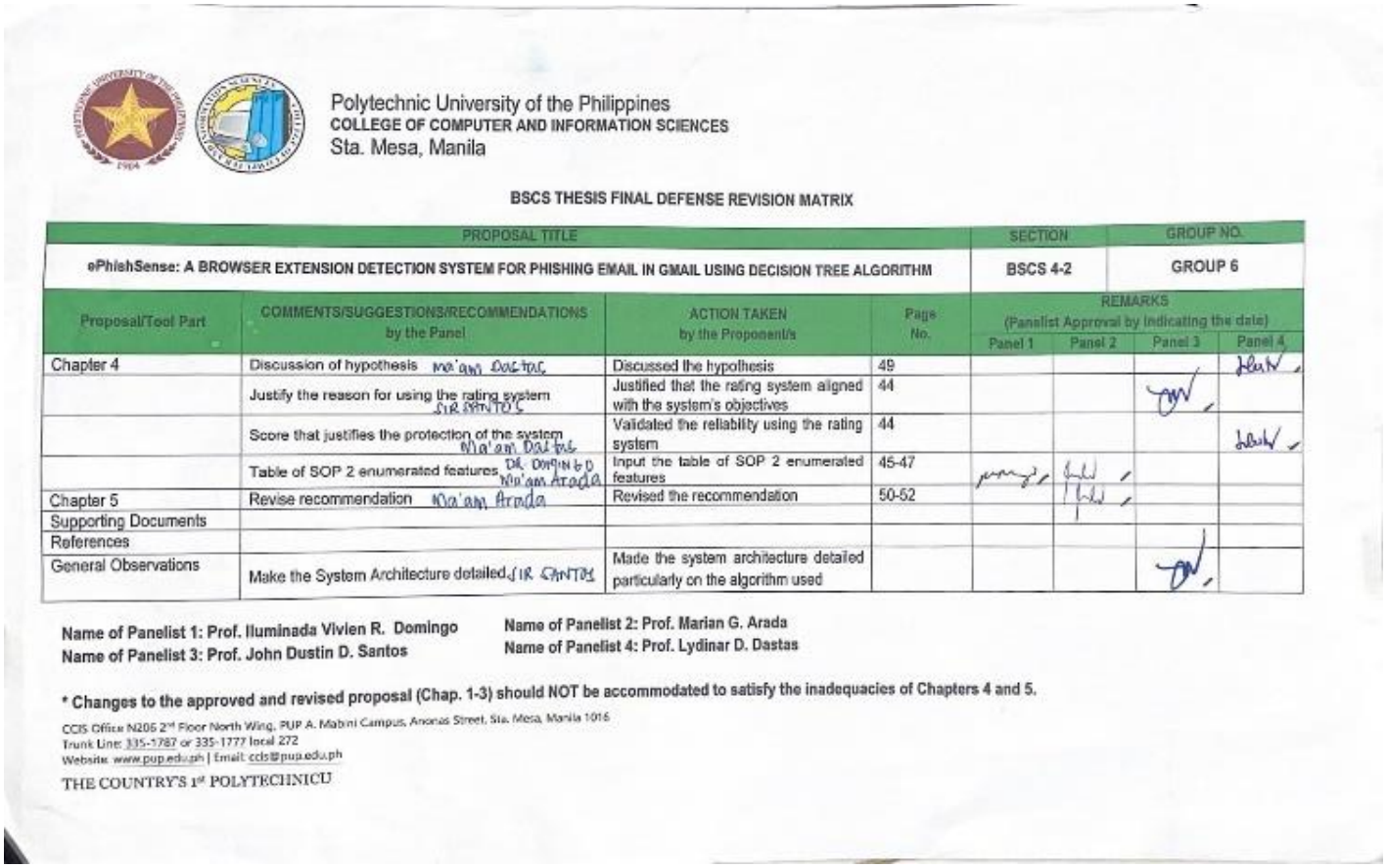
Name of Panelist 1: Prof. Iluminada Vivien R. Domingo
Name of Panelist 2: Prof. Marian G. Arada
Name of Panelist 3: Prof. John Dustin D. Santos
Name of Panelist 4: Prof. Lydinar D. Dastas

***Revisions should be minimal in a way that it will not change the initially approved proposal and that proposal changes should NOT be accommodated in order to satisfy the inadequacies of the tool created.**

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Trunk Line: 335-1787 or 335-1777 local 272
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Final Defense Revision Matrix



Polytechnic University of the Philippines
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
Sta. Mesa, Manila

BSCS THESIS FINAL DEFENSE REVISION MATRIX

PROPOSAL TITLE	SECTION	GROUP NO.
ePhishSense: A BROWSER EXTENSION DETECTION SYSTEM FOR PHISHING EMAIL IN GMAIL USING DECISION TREE ALGORITHM	BSCS 4-2	GROUP 6

Proposal/Tool Part	COMMENTS/SUGGESTIONS/RECOMMENDATIONS by the Panel	ACTION TAKEN by the Proponent/s	Page No.	REMARKS (Panelist Approval by indicating the date)			
				Panel 1	Panel 2	Panel 3	Panel 4
Chapter 4	Discussion of hypothesis Ma'am Dastas	Discussed the hypothesis	49				
	Justify the reason for using the rating system SIR SANTOS	Justified that the rating system aligned with the system's objectives	44				
	Score that justifies the protection of the system Ma'am Dastas	Validated the reliability using the rating system	44				
	Table of SOP 2 enumerated features DR. DOMINGO Ma'am Arada	Input the table of SOP 2 enumerated features	45-47				
Chapter 5	Revise recommendation Ma'am Arada	Revised the recommendation	50-52				
Supporting Documents							
References							
General Observations	Make the System Architecture detailed SIR SANTOS	Made the system architecture detailed particularly on the algorithm used					

Name of Panelist 1: Prof. Iluminada Vivien R. Domingo
Name of Panelist 2: Prof. Marian G. Arada
Name of Panelist 3: Prof. John Dustin D. Santos
Name of Panelist 4: Prof. Lydinar D. Dastas

*** Changes to the approved and revised proposal (Chap. 1-3) should NOT be accommodated to satisfy the inadequacies of Chapters 4 and 5.**

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